Cellular Adaptation
&
Reversible Cell Injury

Prof Dr Mohammad Zulkarnaen

MB BCh BAO, MPath, PgDipT&L, MIAC, AM

Learning objectives

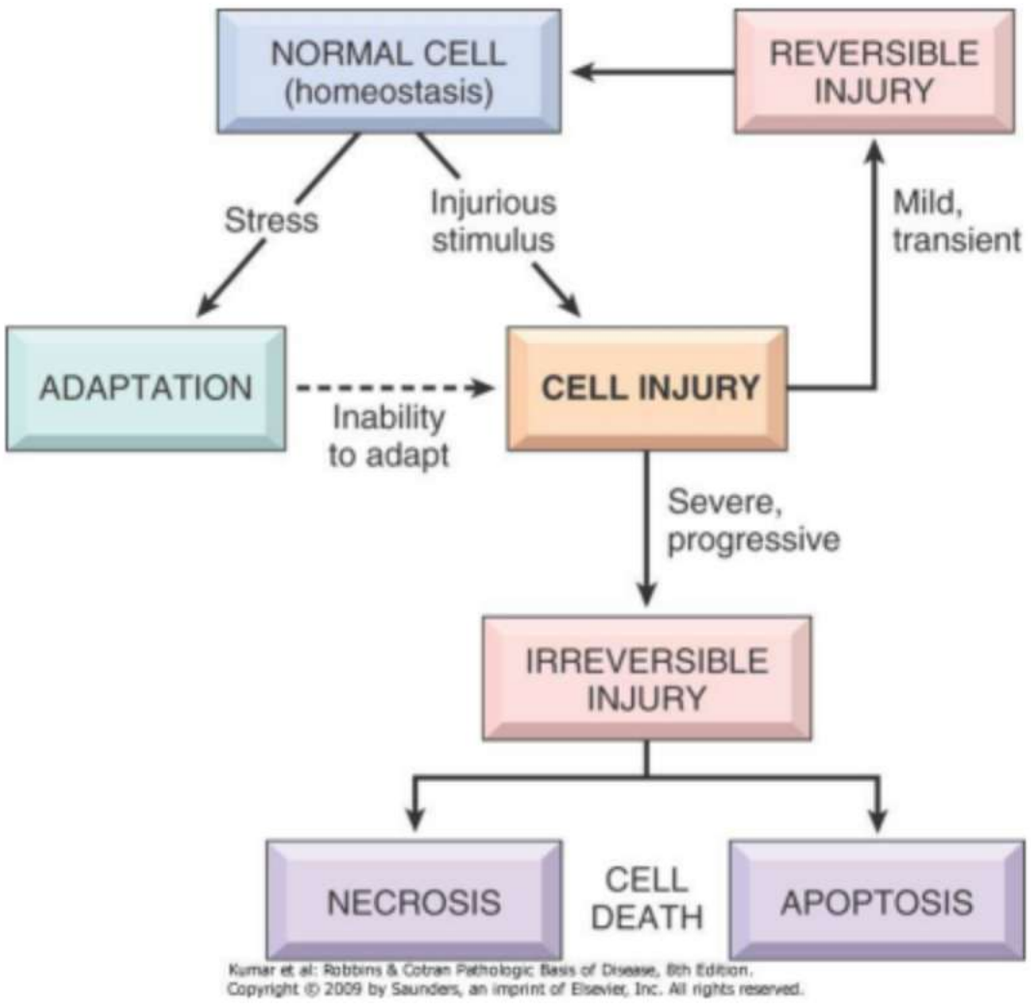
At the end of the teaching session, the student should be able to:

- Describe differences between hyperplasia, hypertrophy, atrophy and metaplasia.
- Explain causes and mechanism of cell injury.
- Describe reversible cell injury: fatty change, hydropic degeneration and briefly about intracellular accumulations.
- Describe different types of intracellular cellular accumulations.

Pathology: the study of the structural, biochemical, and functional changes in cells, tissues, and organs that underlie disease.

- **Core Concepts of Disease (4 Key Aspects)**
- **Aetiology (Cause of disease)**
  – *Why does the disease happen*
  – Can be **genetic** or **environmental** (e.g., infection, toxins, nutrition).
- **Pathogenesis (Mechanism of disease)**
  – *How the disease develops*
  – Steps from the initial cause → to cell injury → to disease.
- **Morphologic Changes**
  – *What the tissues look like under the microscope or grossly*
  – Structural changes due to the disease.
- **Clinical Manifestations**
  – *How the disease presents in the patient*
  – Signs, symptoms, and consequences of the structural changes.

Stages of the cellular response to stress and injurious stimuli.



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Injury to the **CELL**,
is the basis of all diseases.

- **Cells respond to stress in 3 main ways:**
- **Adaptation**
  – Cell changes but survives.
- **Reversible Injury**
  – Mild damage; the cell can recover.
- **Irreversible Injury → Cell Death**
  – Severe damage; the cell cannot recover
  – Leads to **necrosis** or **apoptosis**.
- **Key point:**
- *Cell injury is the basis of most diseases.*

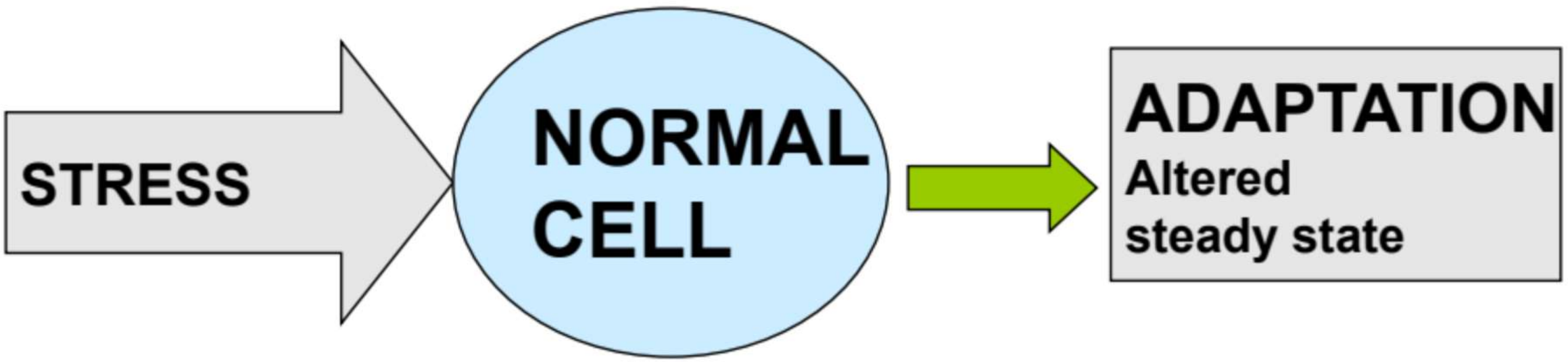
Cellular adaptation

- **Definition** : Cell's ability to change its size, number, or phenotype in response to physiological or pathological stress.

- **Purpose** : To protect the cell from injury.

- **Adapted cell** : Not normal but not injured.

- **Types of adaptation**: Hyperplasia, Hypertrophy, Atrophy and Metaplasia



Hyperplasia | Hypertrophy | Atrophy | Metaplasia

MORE CELLS

Increase in the size of an organ or tissue caused by an increase in the number of cells.

Occurs only in cells capable of dividing (mitotic cells).

1. Physiologic

    * Hormonal (endometrium/breast/uterus in pregnancy)
    * Compensatory (partial hepatectomy, erythrocytosis)

2. Pathologic

Excessive hormone/Growth Factor stimulation of target tissue

    * Endometrial hyperplasia (excess oestrogen) → can lead to malignancy
    * Benign prostatic hyperplasia (excess androgens)
    * Connective tissue cells in wound healing

Hyperplasia | Hypertrophy | Atrophy | Metaplasia



- **MECHANISM**
- Driven by **growth factors** that stimulate mature cells to divide.
- Or by **activation of tissue stem cells** producing new cells.
- Example: After partial hepatectomy, growth factors in the liver stimulate cell proliferation

Hyperplasia | Hypertrophy | Atrophy | Metaplasia

BIGGER CELLS

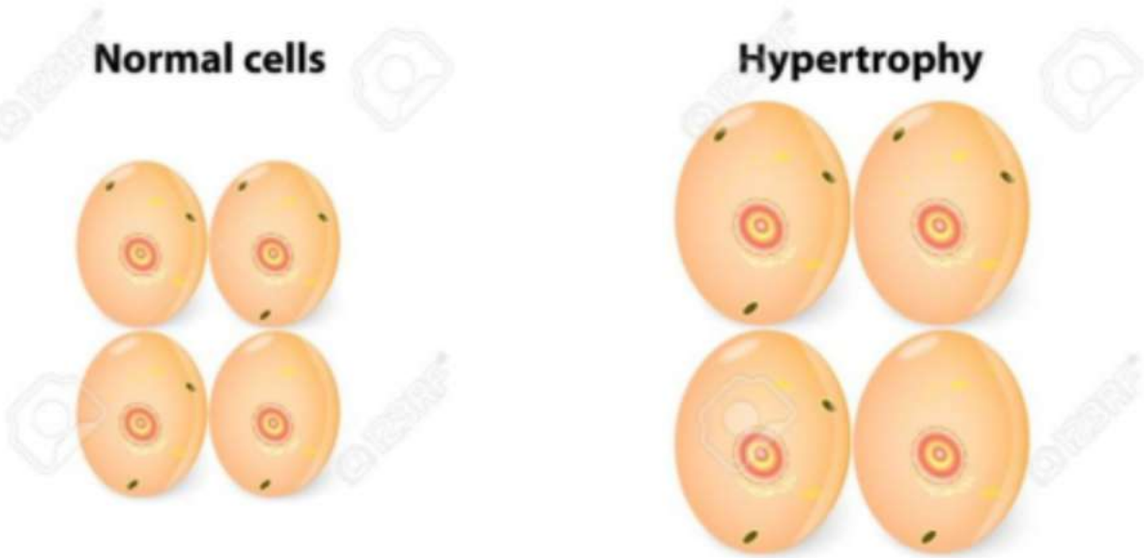
It is an increase in the size of an organ or tissue due to an increase in the <u>size of cell.</u>

permanent cell (cardiac, skeletal) → X hyperplasia

<u>Physiologic</u>
Hypertrophy of skeletal muscle in bodybuilder

<u>Pathologic</u>
Hypertrophy of the heart in hypertension

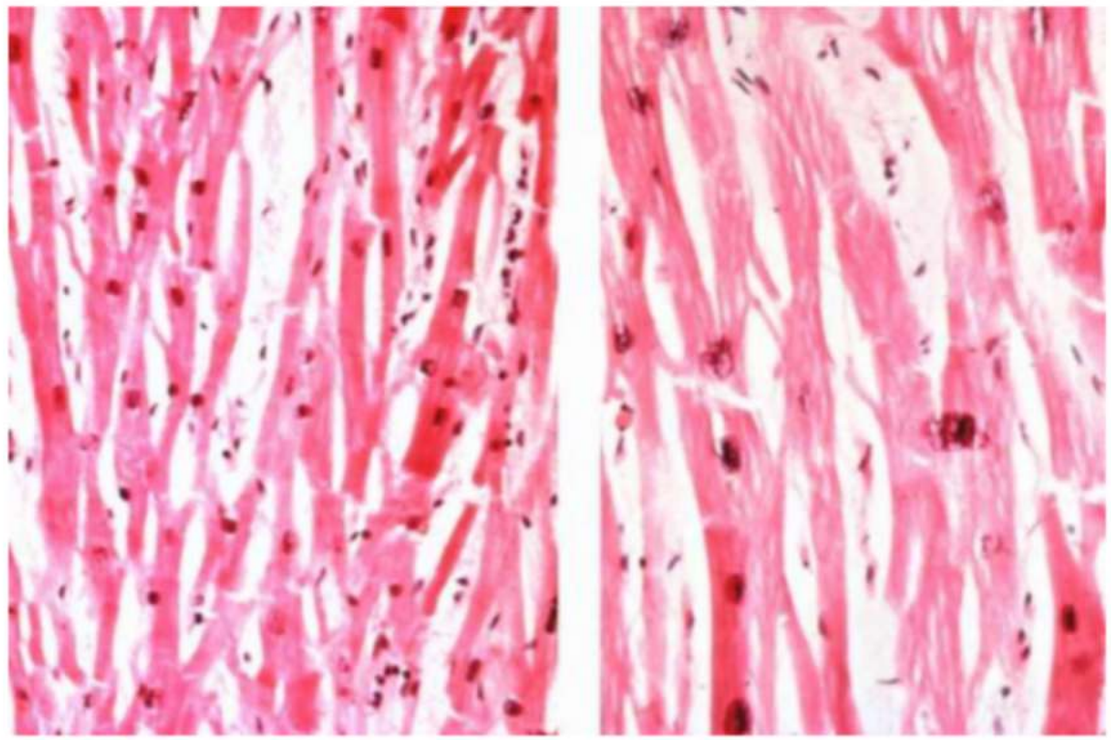
Hyperplasia | **Hypertrophy** | Atrophy | Metaplasia



- Mechanism:
- Increase in the protein synthesis and an increase in the size or number of intracellular organelles.
- End Result is larger organ
- Increase in functional capacity
- * May occur with hyperplasia

Hyperplasia | Hypertrophy | Atrophy | Metaplasia

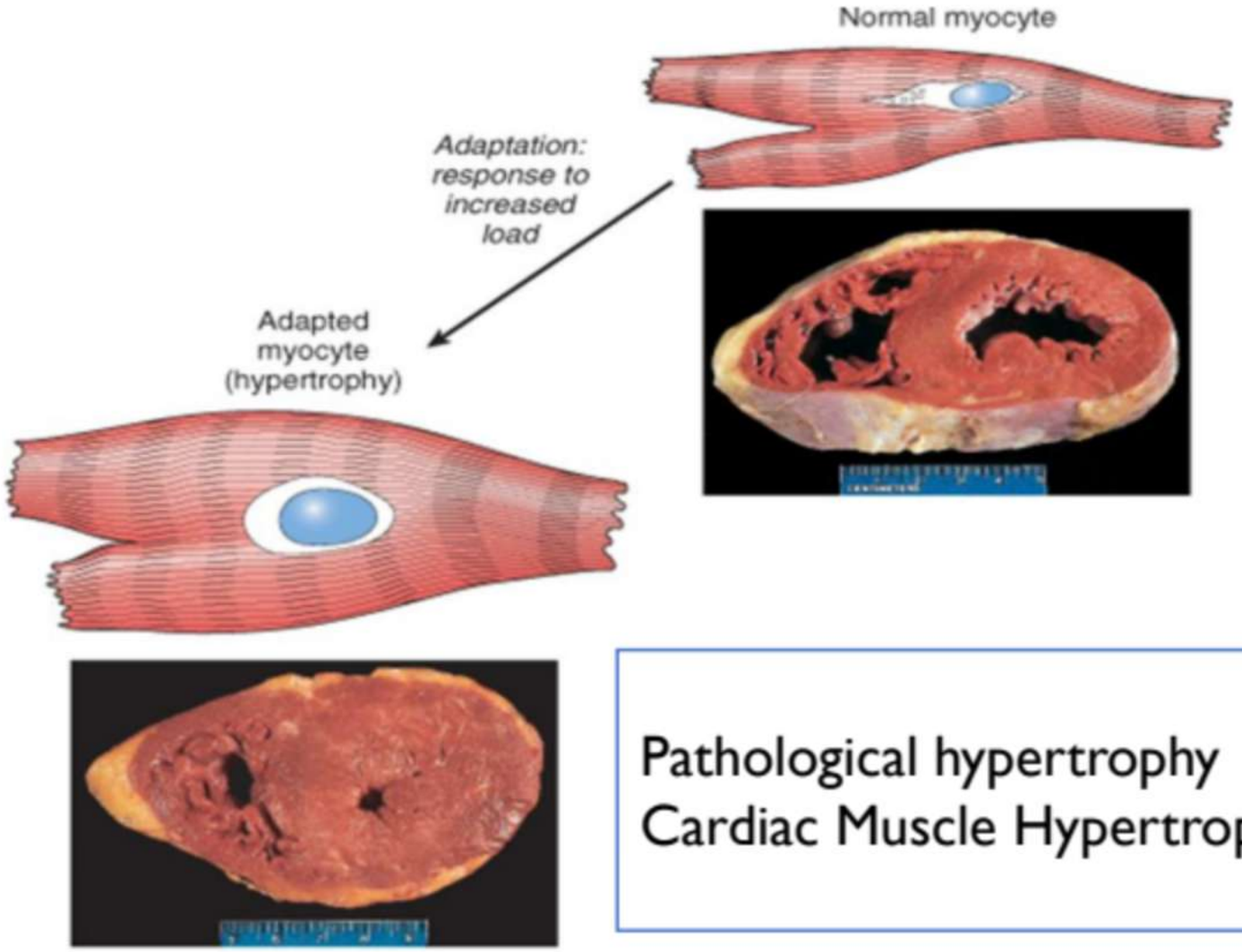




Normal

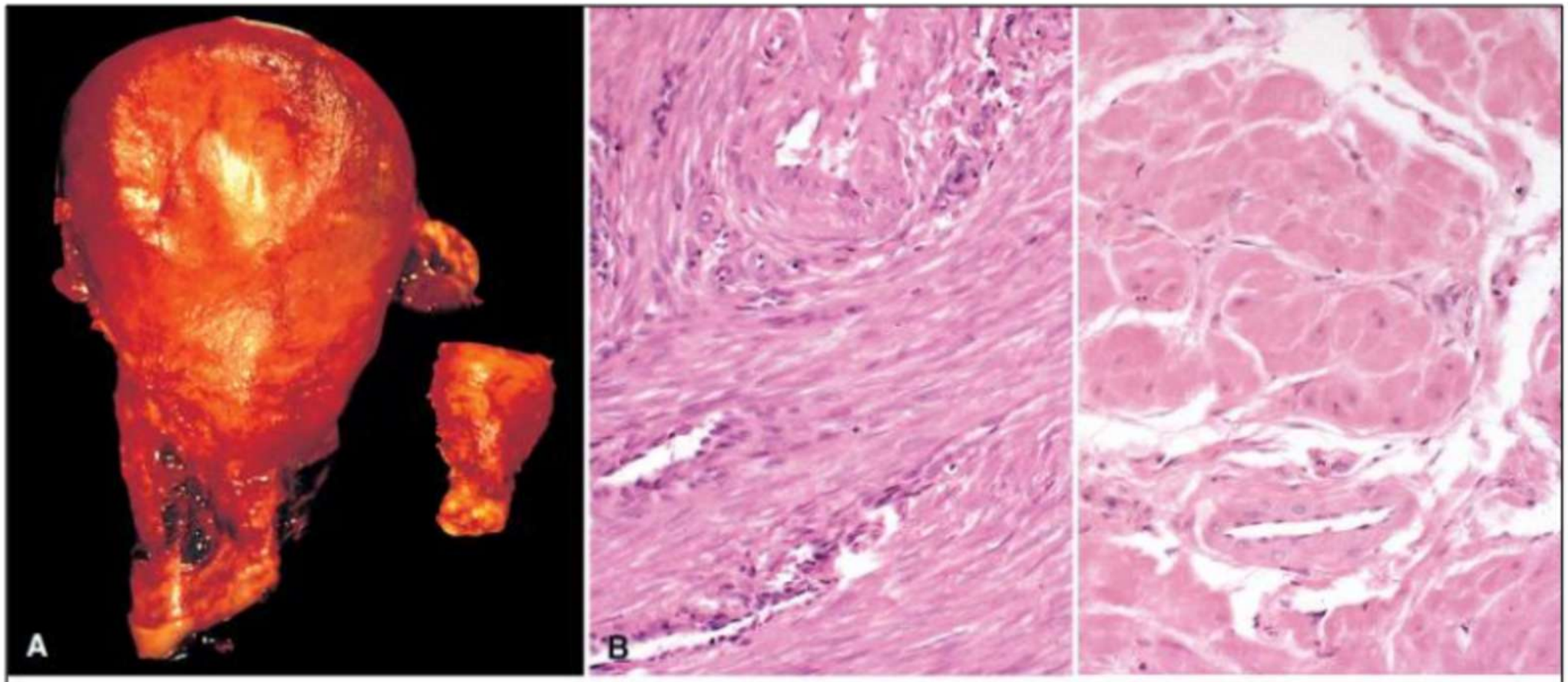
Hypertrophy

Pathologic Hypertrophy in systemic hypertension



Caused by chronic pressure overload (e.g., hypertension).

Hypertrophy and Hyperplasia of Uterus



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- Cells capable of division (labile cells and stable cells) can undergo both hyperplasia and hypertrophy
- *nondividing cells* or permanent cells (e.g., myocardial fibers) can only undergo hypertrophy.

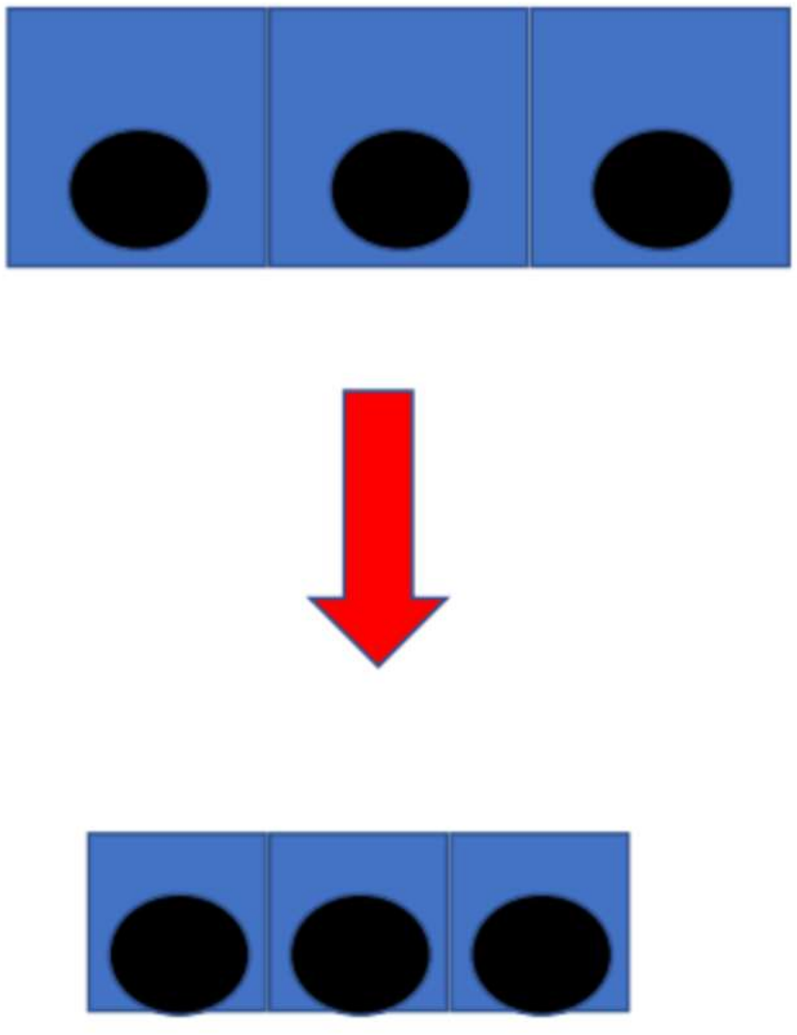
Hyperplasia | Hypertrophy | **Atrophy** | Metaplasia

SMALLER CELLS

Atrophy is decrease in the size of an organ or tissue due to **decrease in cell size and/or cell number**

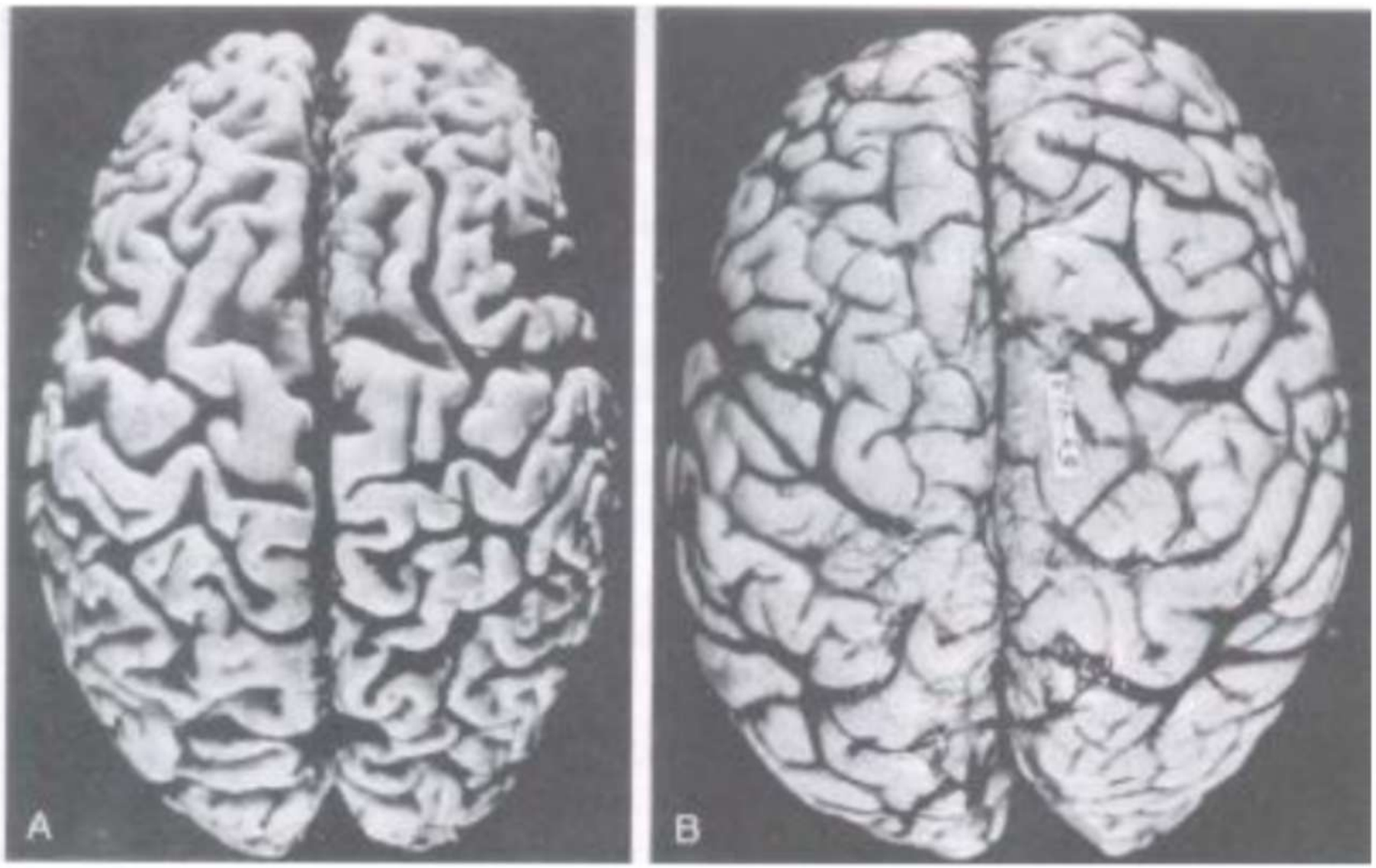
- **Physiological:** thymus, uterus after delivery
- **Pathological:** (local, generalised)
  1. Denervation
  2. Loss of endocrine stimulation
  3. Aging
  4. Disuse atrophy
  5. Pressure – e.g. tumour compression
  6. Ischemia - decrease in blood supply to an organ
  7. Inadequate Nutrition

Hyperplasia | Hypertrophy | **Atrophy** | Metaplasia



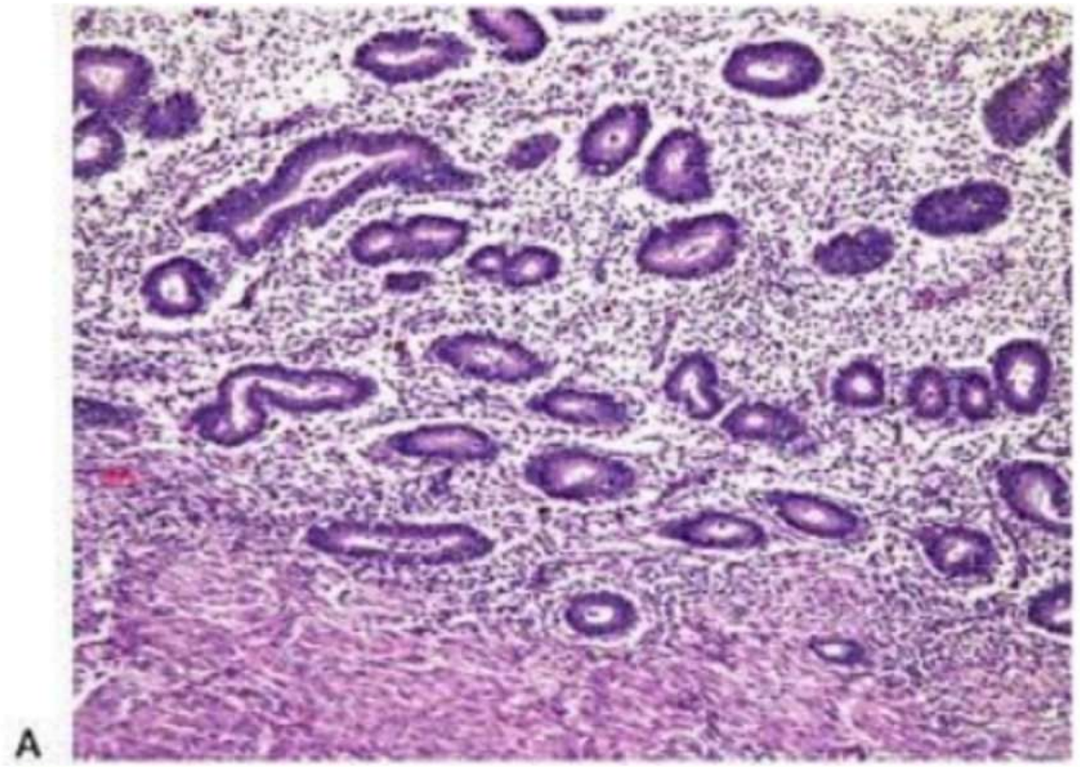
- **Mechanism:**
- **Reduction – cell structural components**
- **The main event in atrophy is <u>degradation of the proteins</u>.**
- **There are two major systems for this:**
- **1. Lysosomes**
- **2. The ubiquitin-proteasome pathway**
- ***Atrophic cell – diminished function but NOT DEAD***

Hyperplasia | Hypertrophy | **Atrophy** | Metaplasia

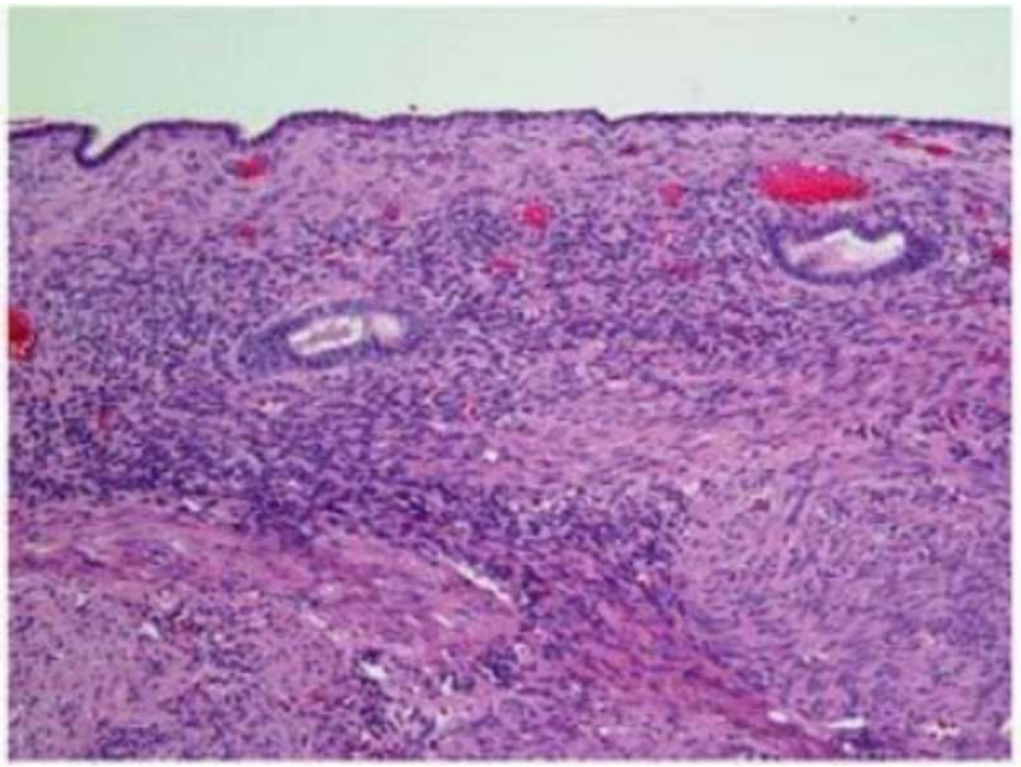


Widened sulci and narrowed gyri

Hyperplasia | Hypertrophy | **Atrophy** | Metaplasia



Woman in reproductive age
Proliferative endometrium



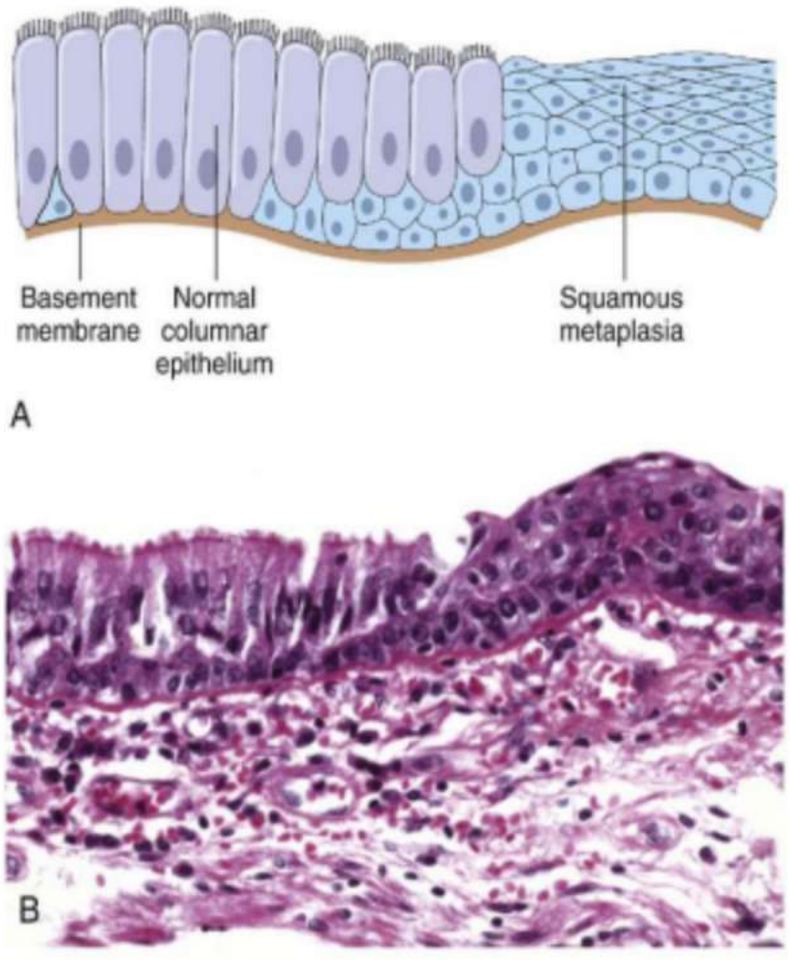
Woman -postmenopausal age: atrophic endometrium

Hyperplasia | Hypertrophy | Atrophy | Metaplasia

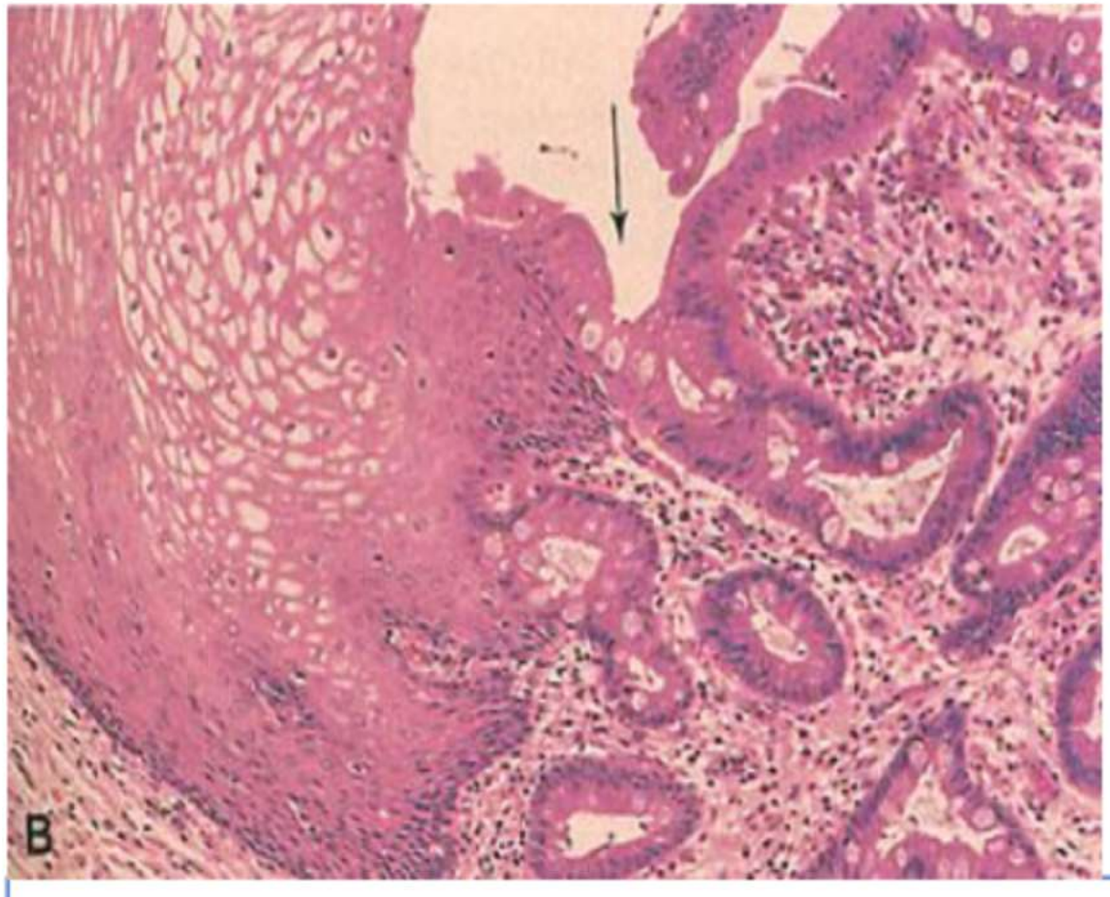
- a reversible change in which one differentiated cell (epithelial or mesenchymal) is replaced by another cell type.

- Fully reversible
- Reprogramming of : STEM cell in epithelia or mesenchymal cell in connective tissue

Metaplasia



Metaplasia of columnar to squamous epithelium in a bronchus
- Seen in chronic smoker



Metaplasia from squamous to columnar epithelium
The oesophageal squamous epithelium has been replaced by metaplastic columnar epithelium, intestinal type. (Barrett esophagus)

Key Concepts : Cellular adaptations to stress

- **Hypertrophy**
- ↑ Cell size → ↑ organ size
- Due to ↑ workload or ↑ growth factors
- Occurs in tissues **without cell division** (e.g., heart)
- **Hyperplasia**
- ↑ Number of cells
- Occurs in tissues that **can divide** or have stem cells
- Physiologic (hormonal) or pathologic

- **Atrophy**
- ↓ Cell size and ↓ organ size
- Due to ↓ nutrients, disuse, denervation, ageing
- ↓ protein synthesis + ↑ protein degradation
- **Metaplasia**
- Change from one adult cell type to another
- Response to chronic irritation (e.g., smoking)
- May reduce function & ↑ risk of malignancy

Cell Injury and Cell Death

Causes of cell injury

- **Ischaemia /Hypoxia.**
- Chemical agents and drugs.
- Mechanical/physical agents.
- Infectious agents: from viruses to parasites.
- Immune effector proteins and cells: e.g. autoimmune disease.
- Growth factor stimulation or removal.
- Nutrient imbalances, specific metabolic inhibitors: protein to vitamin deficiencies, excess cholesterol.

Ischaemia / Hypoxic cell injury

- **Hypoxia = reduced oxygen availability**
  Ischaemia = reduced blood supply (most common cause of hypoxia)

- Causes of hypoxia

- Ischaemia – obstruction of arterial blood flow(most common)
- Anaemia– reduction in oxygen carrying red blood cells
- Carbon-monoxide poisoning– altered haemoglobin.
- Reduced tissue perfusion- cardiac failure, hypotension and shock
- Poor oxygenation of blood – pulmonary disease

- **Key point:** *Ischaemia affects both oxygen delivery **and** nutrient supply, so it causes injury faster than hypoxia alone.*

Chemical agents and drugs

❑ Causes

- Simple chemicals (e.g. glucose / salt in hypertonic concentrations).
- High concentration of oxygen.
- Poisons : arsenic, cyanide, or mercuric salts.
- Alcohol and therapeutic drugs.
- Environmental/occupational toxins: CO, asbestos, insecticides, herbicides

❑ Mechanism

- **1. Direct toxicity**
- Acts on cell membranes or organelles
- Examples: mercury, bacterial toxins, cytotoxic drugs
- **2. Indirect toxicity (via metabolic activation)**
- Formation of **reactive metabolites** → ROS → lipid peroxidation
- Examples: carbon tetrachloride (CCl_4), acetaminophen overdose, cyclophosphamide

Physical agents

❑ Causes

- Mechanical trauma
- Extremes of temperature (burns & deep cold)
- Sudden changes in atmospheric pressure
- Radiation
- Electric shock

Infectious agents

❑ Causes

- Viruses
- Bacteria
- Fungi
- Higher forms of parasites

- **Immunologic Reactions — Causes**
- Although protective, **immune responses can injure cells**.
- **Autoimmune reactions**, allergies, and reactions to microbes/toxins can cause cell injury.

- **Genetic Derangements — Causes**
- **Inherited disorders** such as sickle cell anemia cause abnormal cell structure or function.
- **Enzyme defects** (inborn errors of metabolism) → toxic buildup → cell injury.
- **DNA damage or misfolded proteins** → triggers cell death when irreparable.

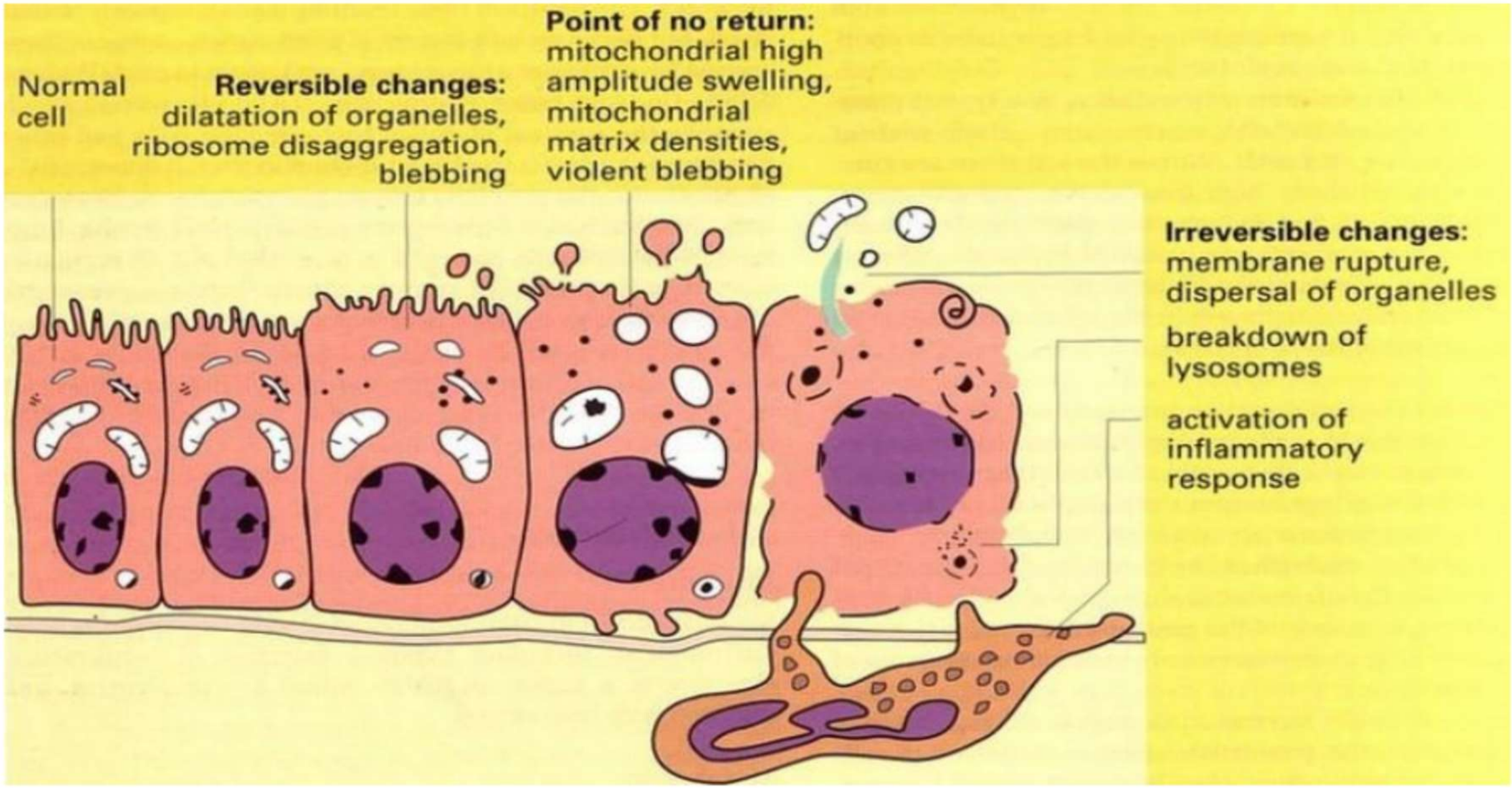
Nutritional imbalances

- **Nutritional Imbalances — Causes**
- **Protein-calorie malnutrition**
- **Vitamin deficiencies** (e.g., A, D, B-complex)
- **Excess nutrients**:
  – High cholesterol → atherosclerosis
  – Obesity → diabetes, fatty liver, certain cancers

Cell Injury – Key Principles

- **Response depends on the type, duration, and severity** of the injury.
- **Outcome depends on the type, state, and adaptability** of the affected cell.
- **Cell injury arises from multiple biochemical mechanisms** affecting essential cellular components.

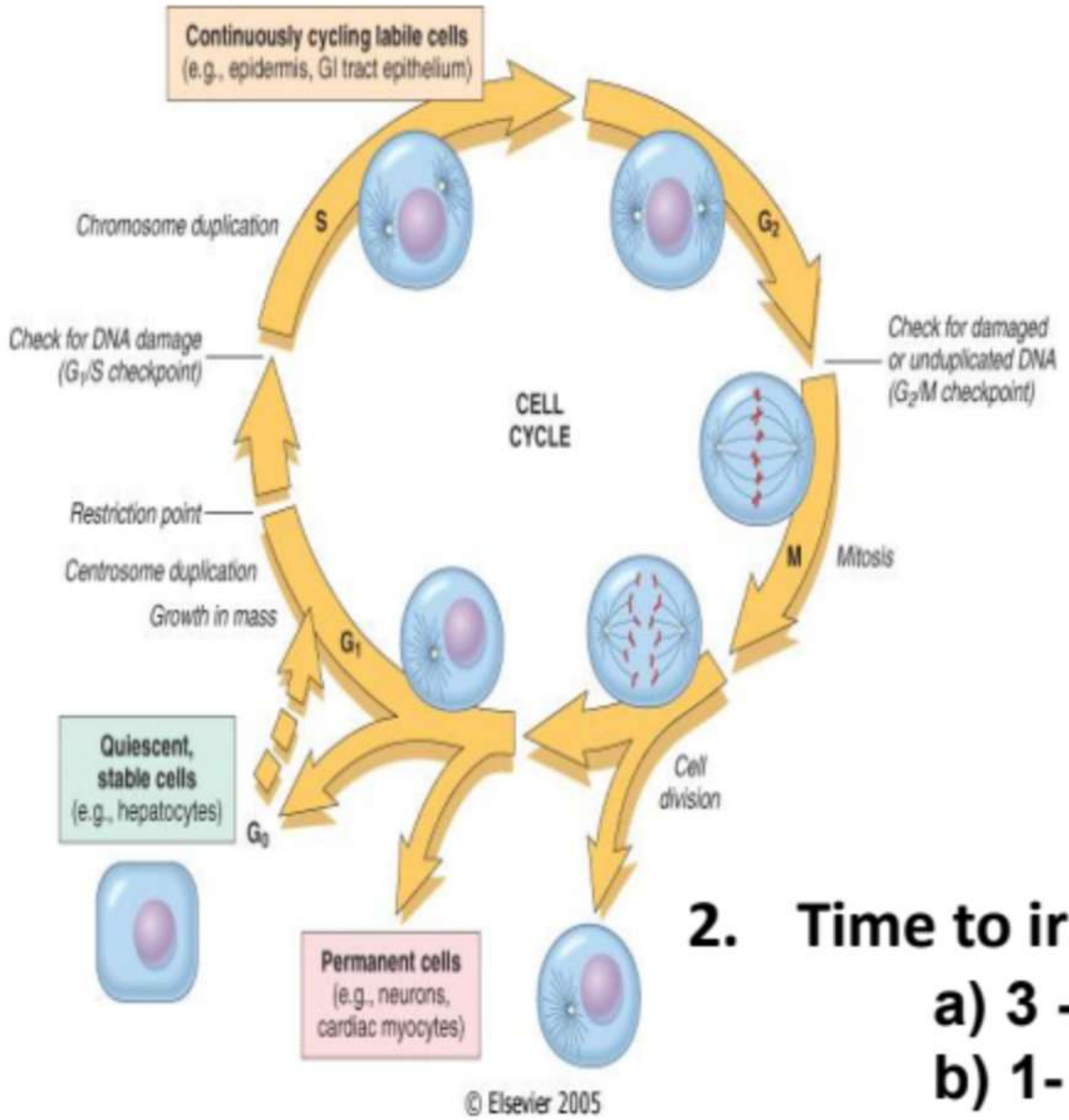
1. The cellular response to injurious stimuli depends on the type of injury, its duration and its severity.



if the injurious stimulus is persistent or severe, the cell suffers irreversible injury

If the injurious stimulus is transient or mild, the cells suffers reversible injury

2) The consequences of cell injury depend on the type, state, and adaptability of the injured cell.



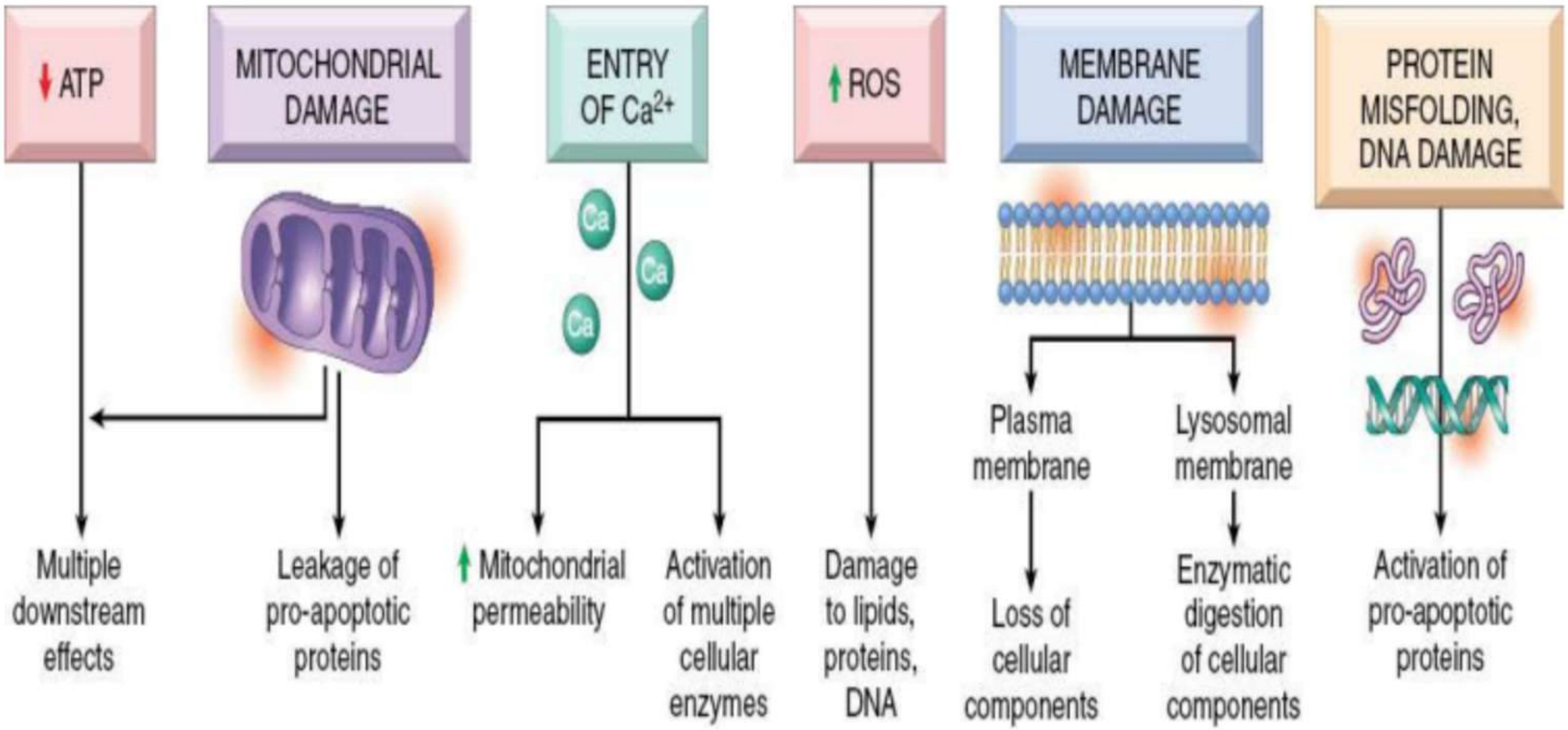
Cell type and Adaptability

- ❑ Labile cells
  - continuously dividing
  - E.g. epithelium, bone marrow
- ❑ Stable cells
  - capable of regeneration following injury
  - E.g. hepatocytes, vascular endothelial cells, osteocytes, chondrocytes.
- ❑ Permanent cells
  - Cannot divide after birth
  - E.g. neurons and cardiac muscle

2. Time to irreversible injury (hypoxia)

a) 3 - 5 min for neurons,
b) 1- 2 hrs. for myocardial & liver cells,
c) many hours for skeletal muscle cells

3) Cell injury results from different biochemical mechanisms acting on several essential cellular components



Cellular injury
Mechanism & Causes

Mechanism of Cell Injury

Mitochondria → energy failure
Membranes → leakage
DNA → apoptosis

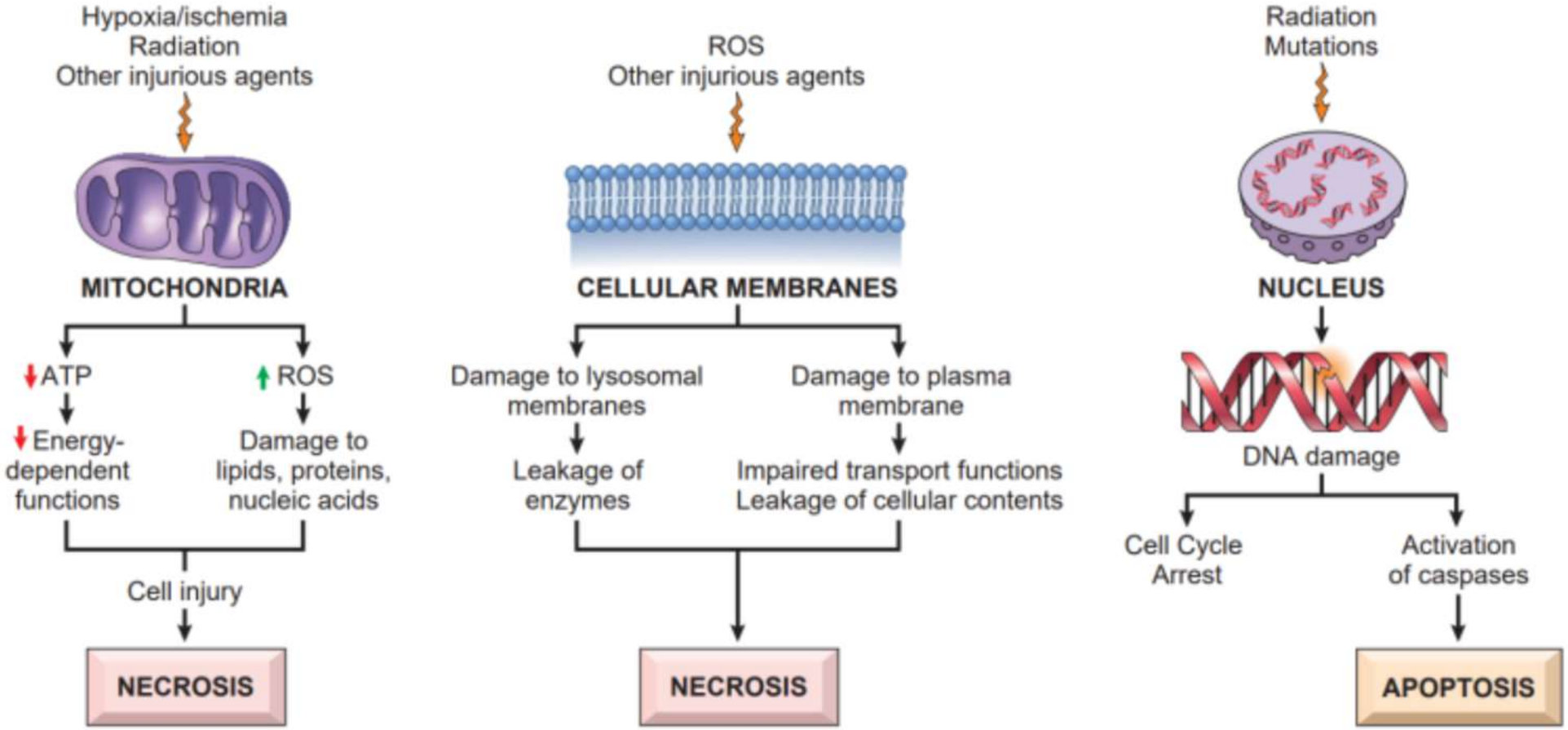
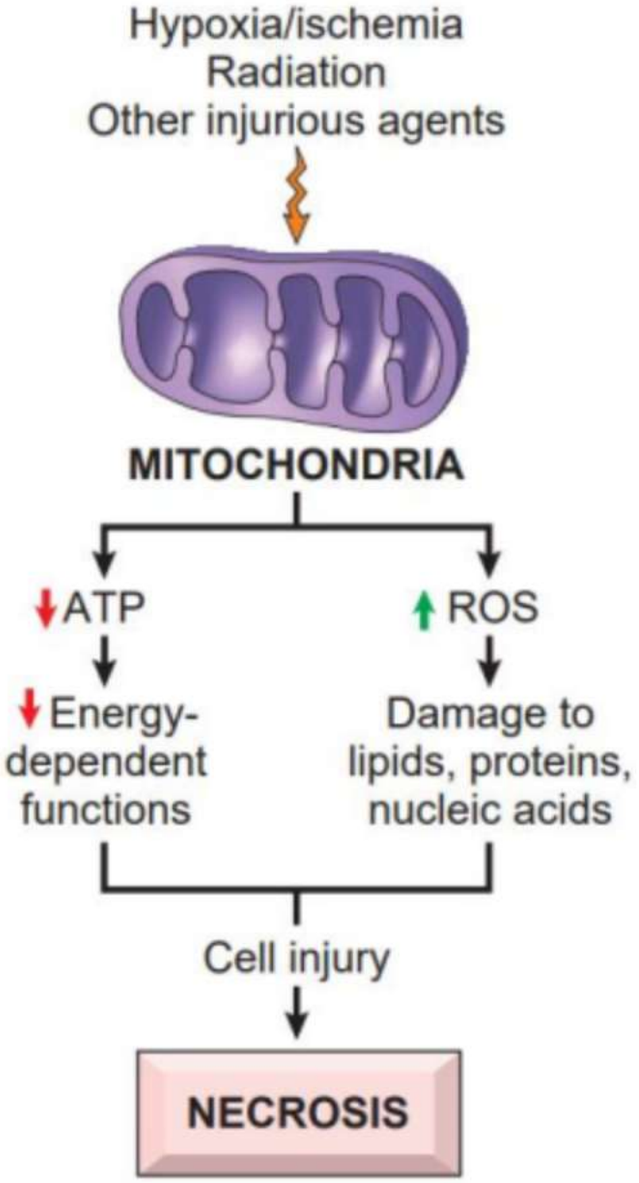
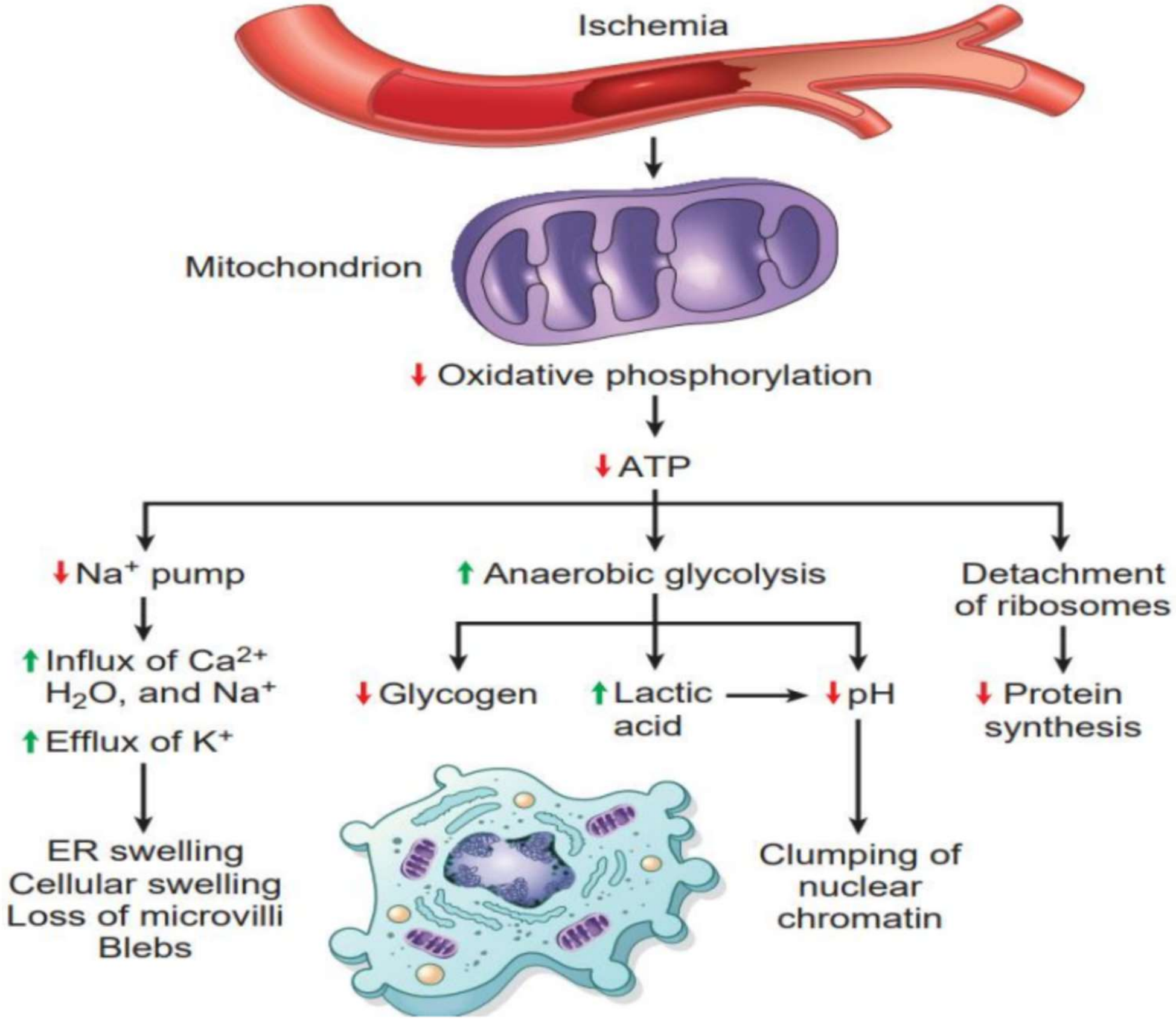


Figure 2.18 The principal forms and sites of damage in cell injury. *ATP*, Adenosine triphosphate; *ROS*, reactive oxygen species.

principal targets of injurious stimuli are mitochondria, cell membranes, protein synthesis machinery, and DNA.



- Mitochondria are critical players in all pathways leading to cell injury and death.
- Mitochondria can be damaged by increases of cytosolic **Ca2+ , ROS, and oxygen deprivation**, which makes them sensitive to virtually all types of injurious stimuli, including hypoxia and toxins.


Ischemia
Mitochondrion
↓ Oxidative phosphorylation
↓ ATP
↓ Na+ pump
↑ Influx of Ca2+ H2O, and Na+
↑ Efflux of K+
ER swelling
Cellular swelling
Loss of microvilli
Blebs
↑ Anaerobic glycolysis
↓ Glycogen
↑ Lactic acid
↓ pH
Clumping of nuclear chromatin
Detachment of ribosomes
↓ Protein synthesis

Ischaemic/Hypoxic cell injury……..contd.

A. Early stage – first mitochondria is affected leading to decreased oxidative phosphorylation and ATP synthesis.

Consequences of Decreased ATP availability:

1. **<u>Failure of cell membrane pump</u>** – intracellular Na + H_2O accumulation, cellular swelling and swelling of organelles. Ca2+ influx

a) **Hydropic** <u>change</u> – large vacuoles in the **<u>cytoplasm</u>**

b) **<u>Swelling of Endoplasmic reticulum</u>** - reversible

c) **<u>Swelling of mitochondria</u> –** reversible to irreversible

Ischaemic/Hypoxic cell injury……contd.

2. Disaggregation of **Ribosomes** – failure of protein synthesis
3. Stimulation of **phosphofructokinase activity** – increased glycolysis, accumulation of lactate and decreased intracellular pH. Acid causes reversible clumping of nuclear material.

B. Late stage –

1. hypoxic injury causes **membrane damage** (plasma membrane, lysosomal, organelle) **with loss of phospholipids**
2. Reversible morphologic signs: myelin figures (damaged cell membrane) and cellular blebing

Ischaemic/Hypoxic cell injury……..contd.

C. Cell death – caused by severe or prolonged injury

1. Irreversible membrane damage → massive Ca^{2+} influx → mitochondrial failure → cell death.

2. **Release** of **intracellular enzymes** and **proteins** from necrotic cells into the circulation (Lab test, indicators of necrosis e.g. myocardial & liver enzymes) Myocardial: Troponin I, CK-MB; Liver: ALT, AST

3. **Vulnerability of cells** to hypoxic injury varies – depending on the **cell type**

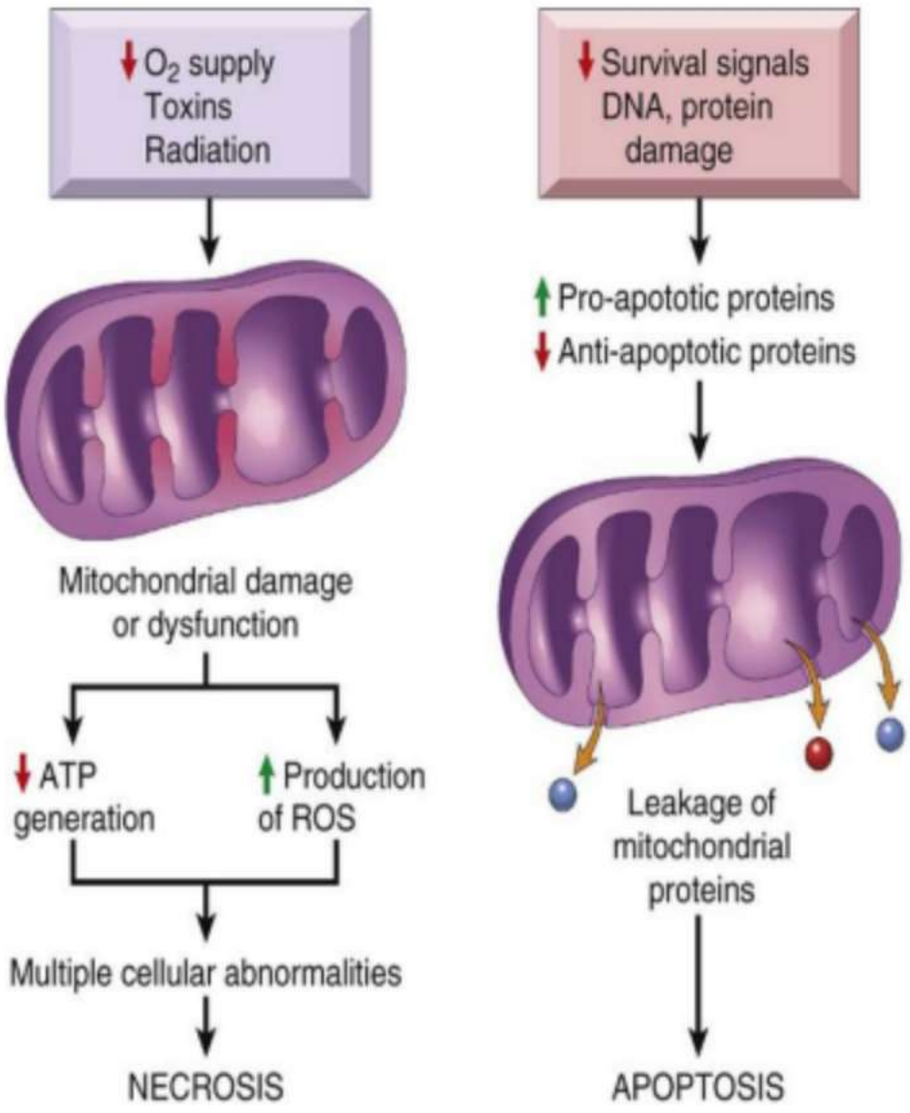
4. **Hypoxic injury becomes irreversible after:**

Neurons (3–5 min) → Myocardium (1–2 h) → Skeletal muscle (many hours)

ISCHAEMIA REPERFUSION INJURY

- **Definition**
- When ischemic but *reversibly injured* cells are reperfused, the restored blood flow can paradoxically worsen the injury.
- **Why it happens**
- Reperfusion exposes vulnerable cells to:
  - **Sudden calcium influx** → mitochondrial damage
  - **Burst of reactive oxygen species (ROS)**
    - from damaged mitochondria
    - from infiltrating neutrophils
- **Inflammation**
  - reperfusion brings neutrophils → more ROS & enzymes
- **Mitochondrial permeability transition (MPT)**
  - leads to ATP depletion → necrosis
- **Clinical importance**
- Major contributor to injury in:
  - **Myocardial infarction**
  - **Cerebral infarction**
- **Take-home point:**
  Reperfusion is necessary to save tissue, but if cells are borderline viable, sudden restoration of blood flow can accelerate oxidative and inflammatory damage.

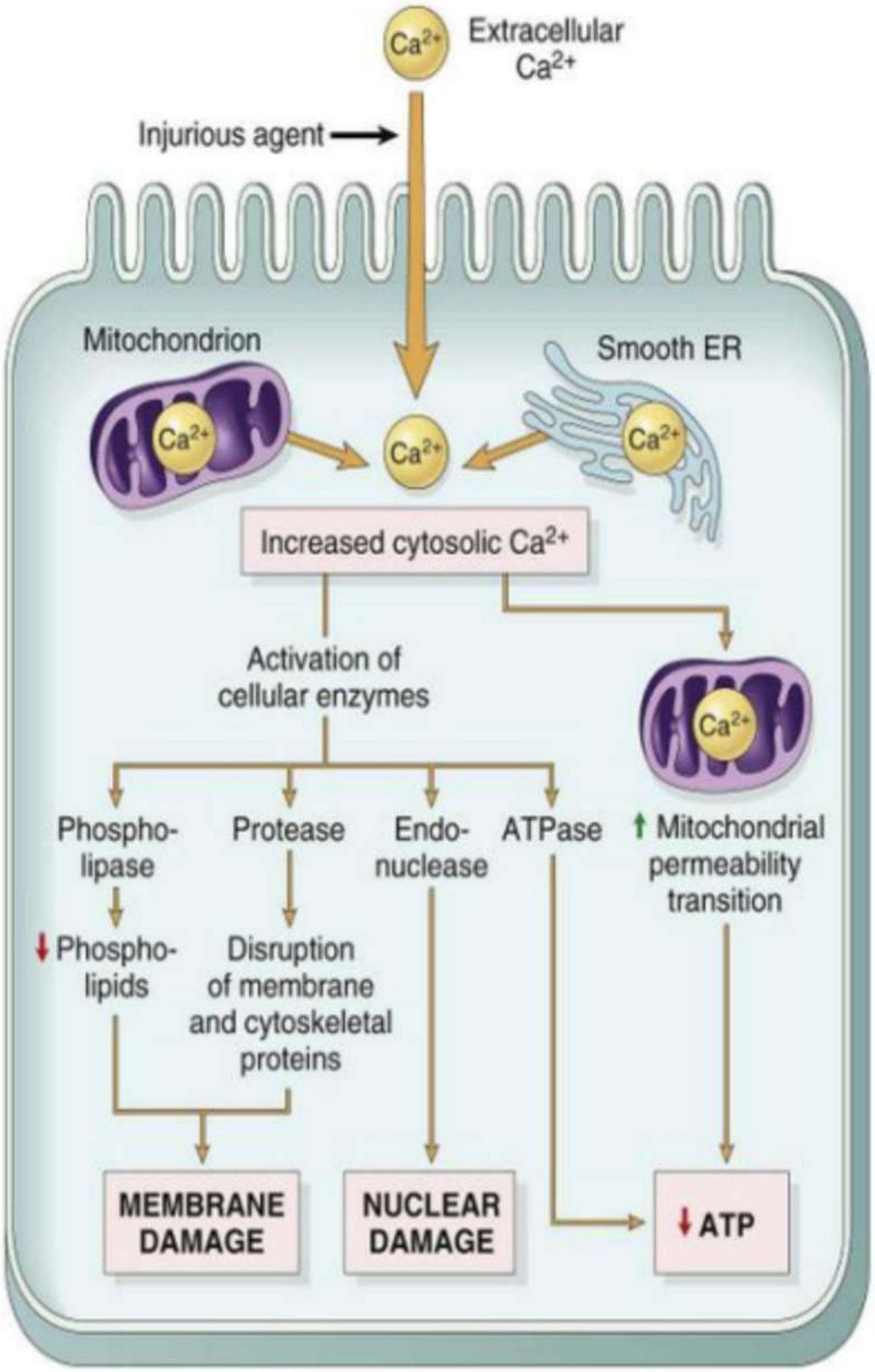
Mitochondrial Damage



- **Consequences of mitochondrial damage:**
- **Formation of the mitochondrial permeability transition (MPT) pore**
  → loss of membrane potential
  → ↓ ATP production
  → **necrosis**
- **Abnormal oxidative phosphorylation**
  → ↑ production of **reactive oxygen species (ROS)**
  → further cell injury
- **Release of pro-apoptotic proteins** (e.g., cytochrome c)
  → activation of **apoptotic pathways**

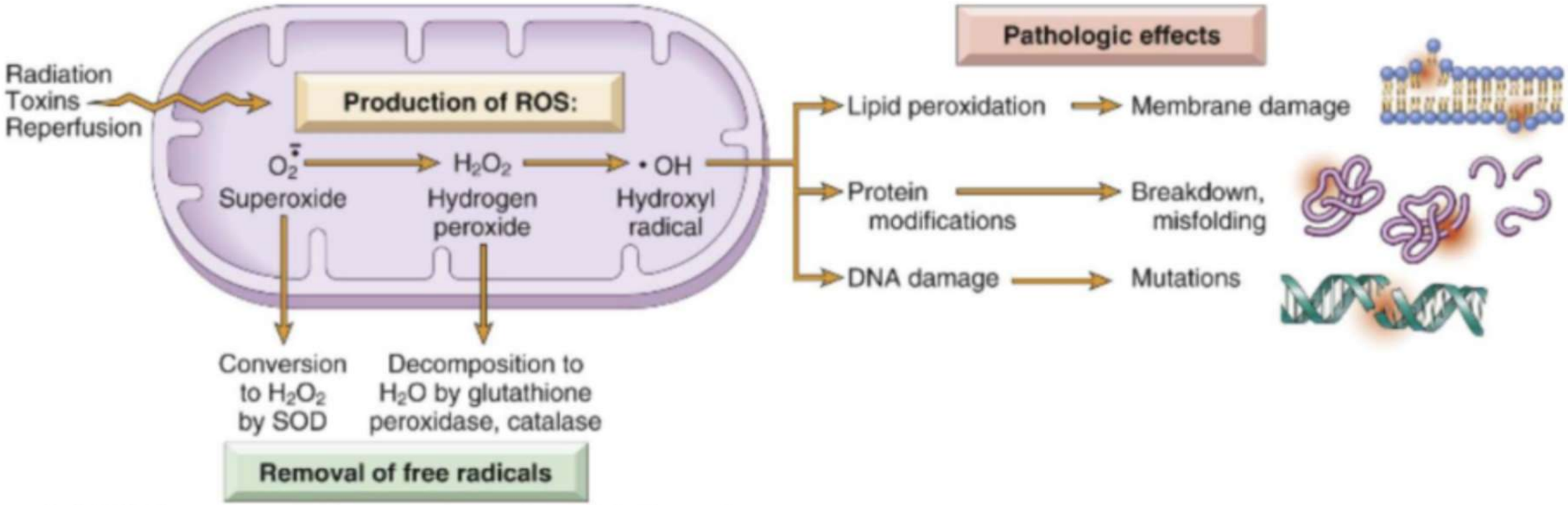
Key idea:
Mitochondrial damage can drive **both necrosis and apoptosis**, depending on the severity and the dominant pathway.

Influx Calcium and loss of Calcium Homeostasis



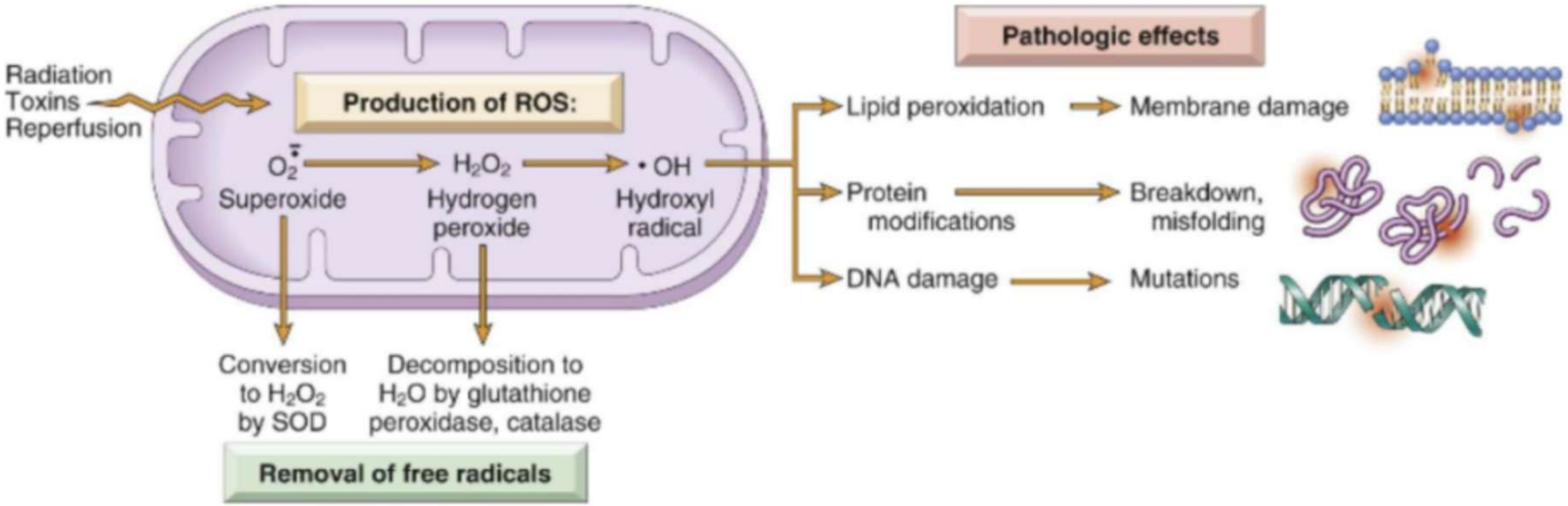
- Cytosolic Ca^{2+} is normally **very low (~0.1 µmol)** compared to extracellular levels **(~1.3 mmol)**.
- Ischemia or toxins → ↑ cytosolic Ca^{2+}
  - first from **ER release**
  - later from **increased influx** across damaged membranes.
- ↑ intracellular Ca^{2+} causes injury by activating enzymes:
  - **phospholipases** → membrane damage
  - **proteases** → cytoskeleton breakdown
  - **endonucleases** → nuclear damage
  - **ATPases** → ATP depletion
- ↑ Ca^{2+} also increases mitochondrial permeability → **apoptosis**.
- **Take-home message:**
  Loss of calcium homeostasis is a major driver of both necrosis (enzyme activation) and apoptosis (mitochondrial injury).

Accumulation of Oxygen-Derived Free Radicals (Oxidative stress)



- ROS are a major cause of cell injury in:
  - chemical & radiation injury
  - ischemia–reperfusion injury
  - Ageing
  - microbial killing by neutrophils/macrophages
- Free radicals = molecules with a single unpaired electron, making them highly reactive.
- They attack nearby molecules including:
  - lipids → membrane damage
  - proteins → misfolding, breakdown
  - DNA → mutations

Accumulation of Oxygen-Derived Free Radicals (Oxidative stress)

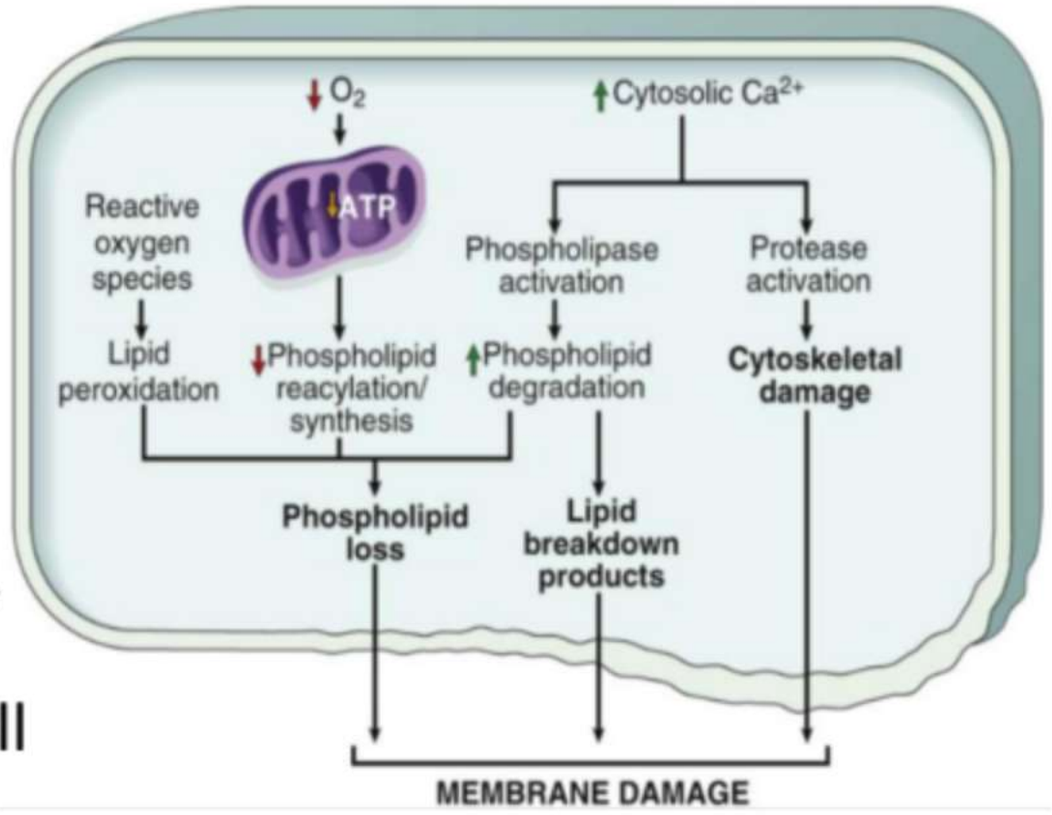


- Reactive oxygen species (ROS) are generated normally during mitochondrial respiration but are usually removed by cellular antioxidant systems.
- At low levels, ROS may be present briefly without causing harm.
- When ROS production exceeds removal, **oxidative stress** occurs.
- Oxidative stress contributes to many diseases, including cell injury, cancer, aging, and neurodegenerative disorders such as Alzheimer's disease.

Defect in Membrane Permeability

- Membrane damage affect the functions and integrity of all cellular membranes.

- **Mechanisms of membrane damage**
  - Reactive oxygen species (lipid peroxidation)
  - Decreased phospholipid synthesis
  - Increased phospholipid breakdown
  - Cytoskeletal abnormalities
- **Pathologic consequences**
  - **Mitochondrial membrane damage** – ↓ ATP, apoptosis
  - **Plasma membrane damage** – Ca^{2+} influx, cell swelling, lysis
  - **Lysosomal membrane injury** – enzyme leakage → necrosis



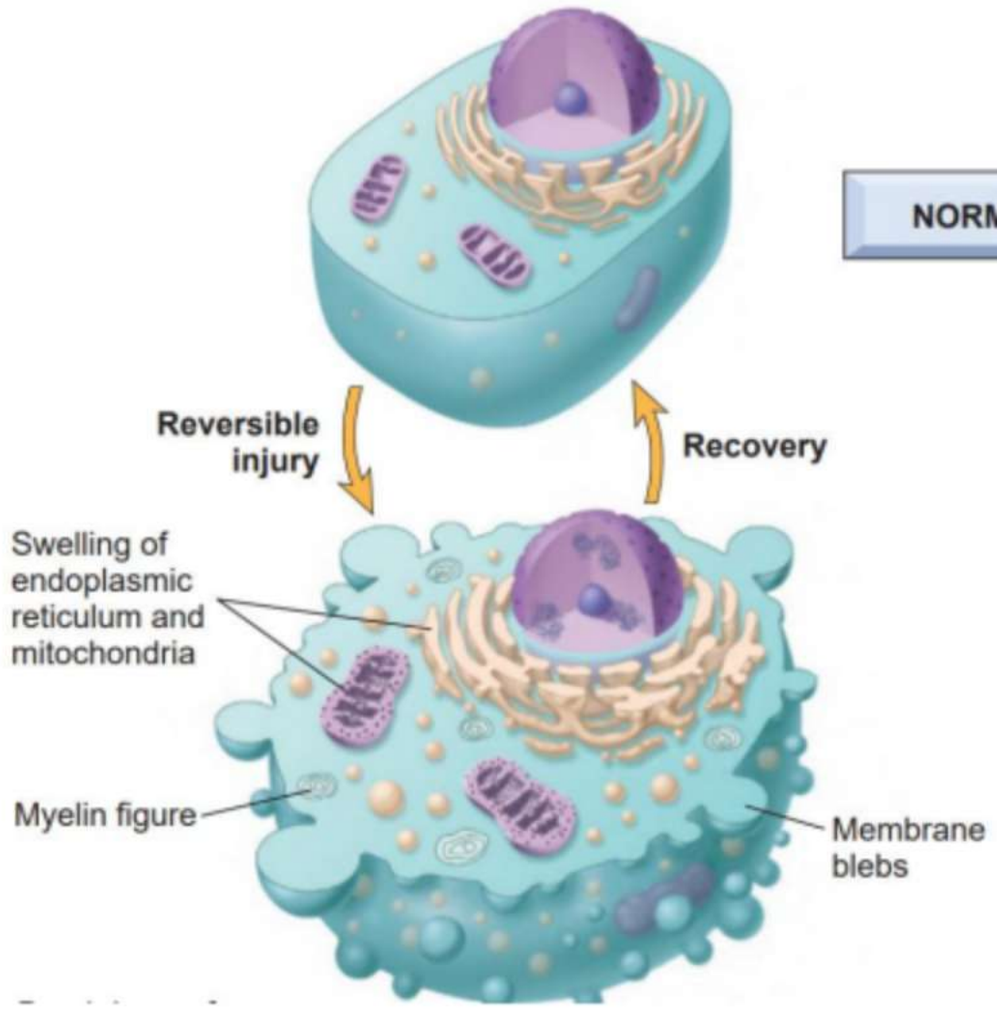
Damage to DNA

- Cells have repair mechanisms to correct DNA damage.
- If the damage is severe (e.g., from radiation, chemotherapy drugs, ROS), repair fails.
- The cell then activates **programmed cell death (apoptosis)** to prevent propagation of mutated DNA.

Key Concepts : Mechanisms of cell injury

- **Depletion of ATP**
  → failure of energy-dependent functions → reversible injury → necrosis
- **Mitochondrial damage**
  → ↓ ATP
  → leakage of pro-apoptotic proteins → apoptosis
  → severe dysfunction → necrosis
- **Calcium influx & loss of calcium homeostasis**
  → activation of enzymes that damage membranes, cytoskeleton, DNA, and proteins
  → may trigger apoptosis
- **Oxidative stress (ROS accumulation)**
  → lipid peroxidation, protein modification, DNA damage
- **Defects in membrane permeability**
  → plasma membrane, mitochondrial membrane, lysosomal membrane damage
- **DNA damage & misfolded proteins**
  → activation of apoptosis pathways

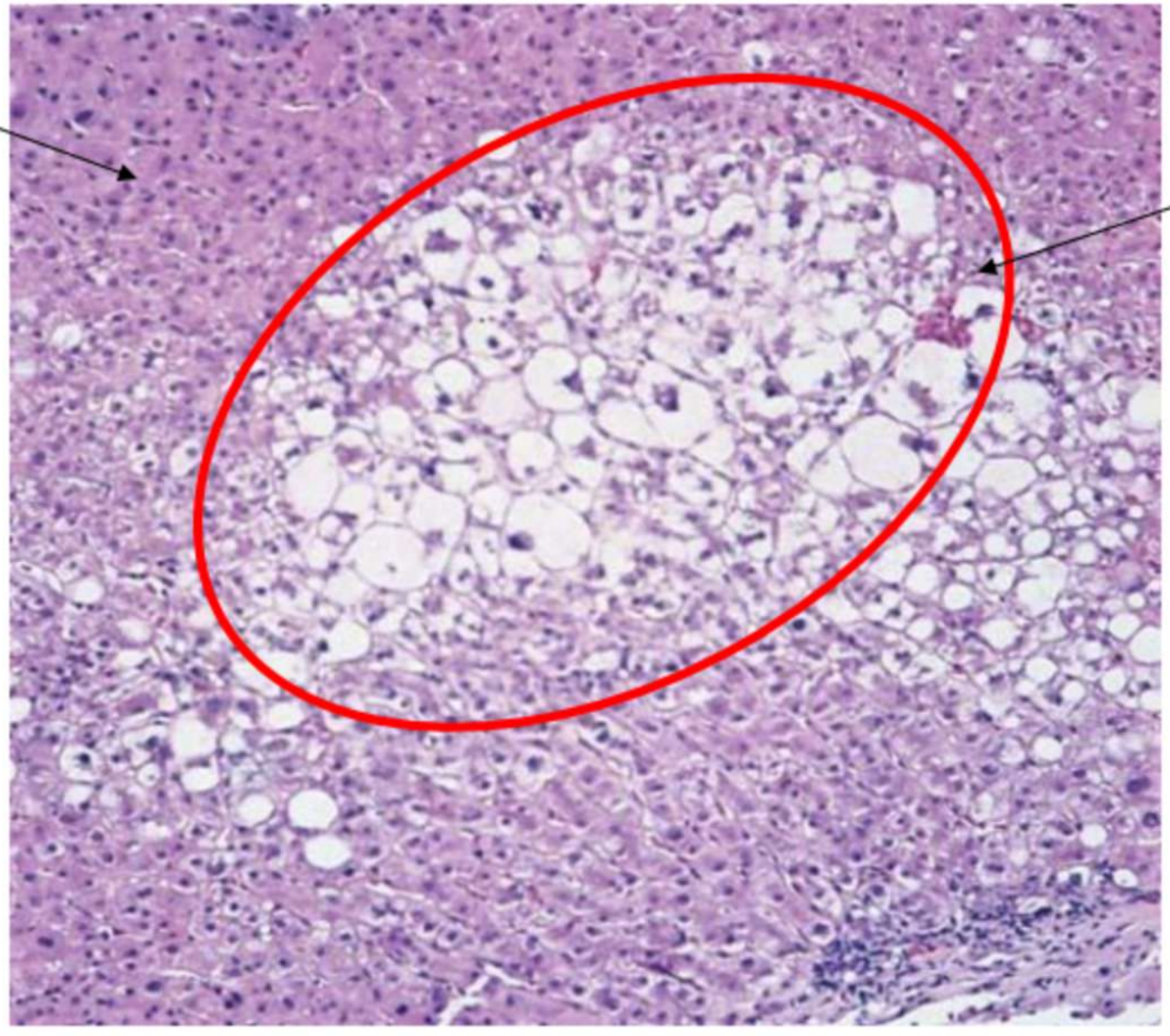
Ultrastructural features of Reversible Cell Injury



- The ultrastructural changes of reversible cell injury, visible by electron microscopy include the following:
- **1. Plasma membrane changes**
  – blebbing, blunting, loss of microvilli
- **2. Mitochondrial swelling**
  – small amorphous densities may appear
- **3. Myelin figures**
  – phospholipid-rich whorls from damaged membranes
- **4. ER swelling**
  – detachment of ribosomes/polysomes
- **5. Nuclear changes**
  – disaggregation of granular and fibrillar components

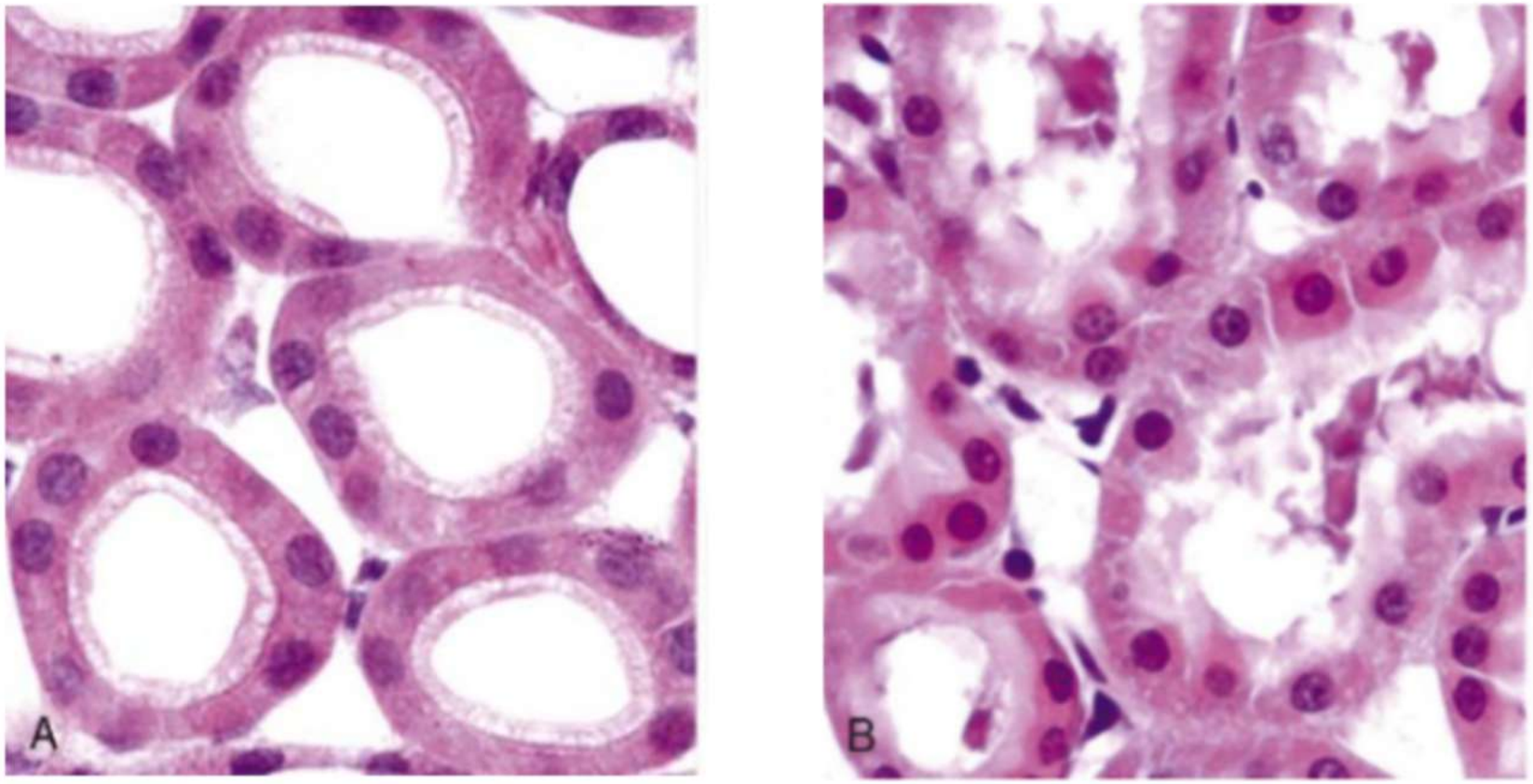
Hydropic swelling - Liver

Normal liver cells



Hydropic change

Morphology of Reversible cell injury



Cellular swelling

A : Normal kidney tubules with viable epithelial cells

B : Early (reversible) ischemic injury showing surface blebs, increased eosinophilia of cytoplasm, and swelling of occasional cells.

Key Concepts : Morphologic Changes in Reversible Cell Injury

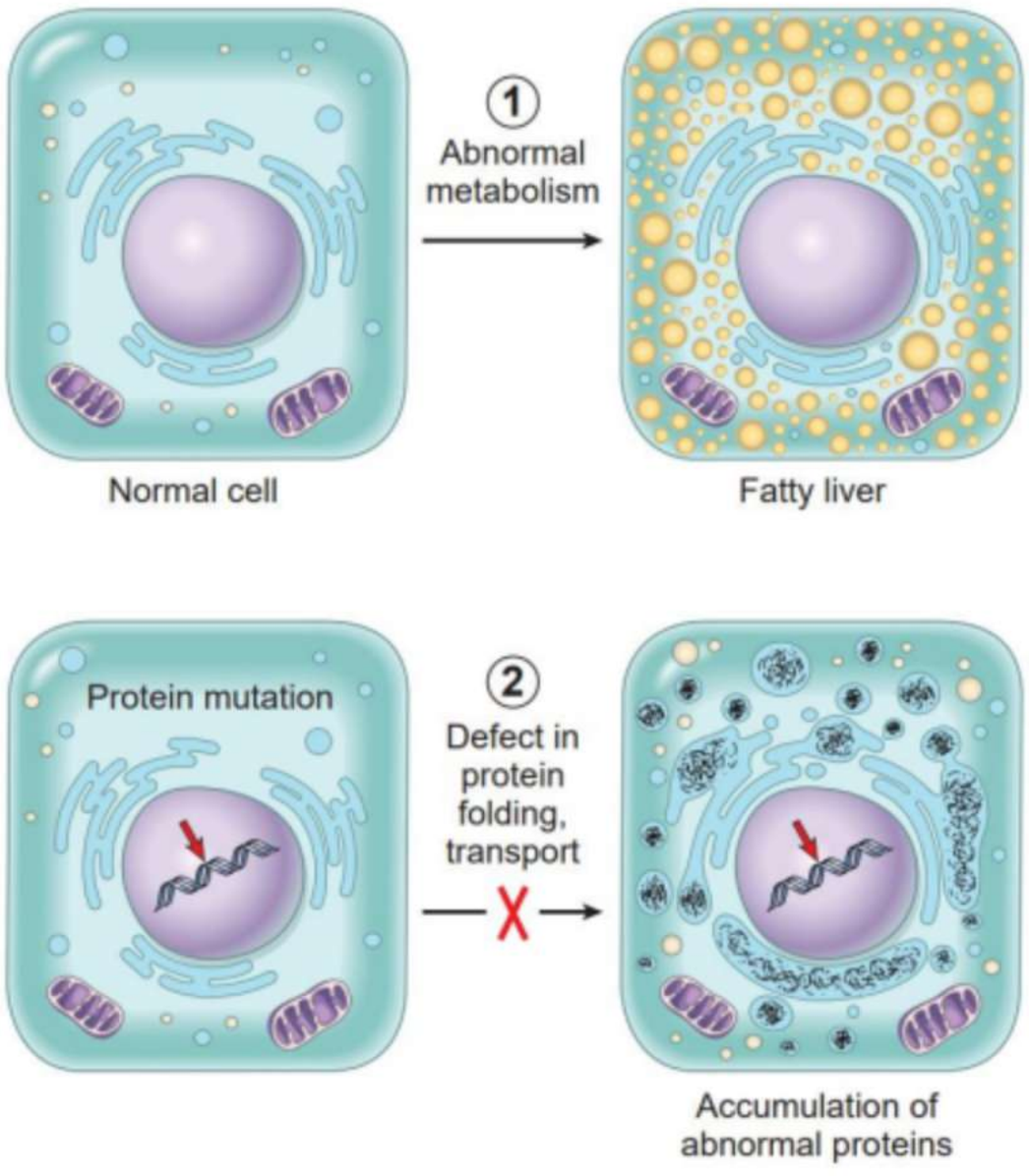
- ❑ **Cellular swelling (hydropic change)**
- ❑ **Fatty change (especially liver, myocardium)**
- ❑ **Plasma membrane blebbing and loss of microvilli**
- ❑ **Mitochondrial swelling**
- ❑ **Dilation of the ER with ribosome detachment**
- ❑ **Mild cytoplasmic eosinophilia (↓ cytoplasmic RNA)**

Intracellular accumulation

Intracellular accumulation

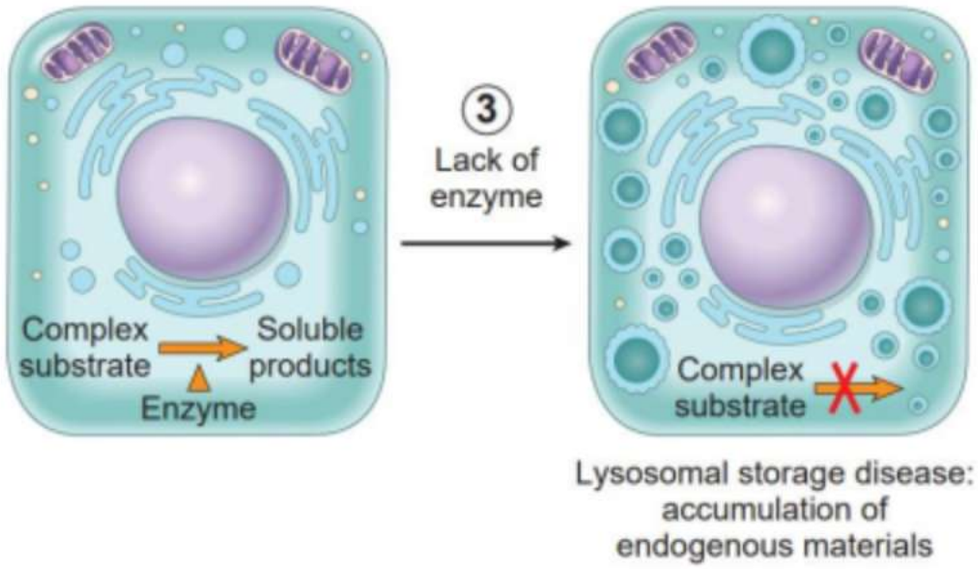
- **Characteristics**
  - Manifestations of metabolic derangements in cells
  - May accumulate transiently or permanently
  - May be harmless or toxic
  - May be located in the cytoplasm or nucleus
- **Types of accumulated substances**
- **Normal cellular constituents:** water, lipids, proteins, carbohydrates
- **Abnormal substances:** minerals, products of infections, misfolded proteins, metabolites
- **Pigments:** endogenous or exogenous

Mechanisms of Intracellular Accumulation

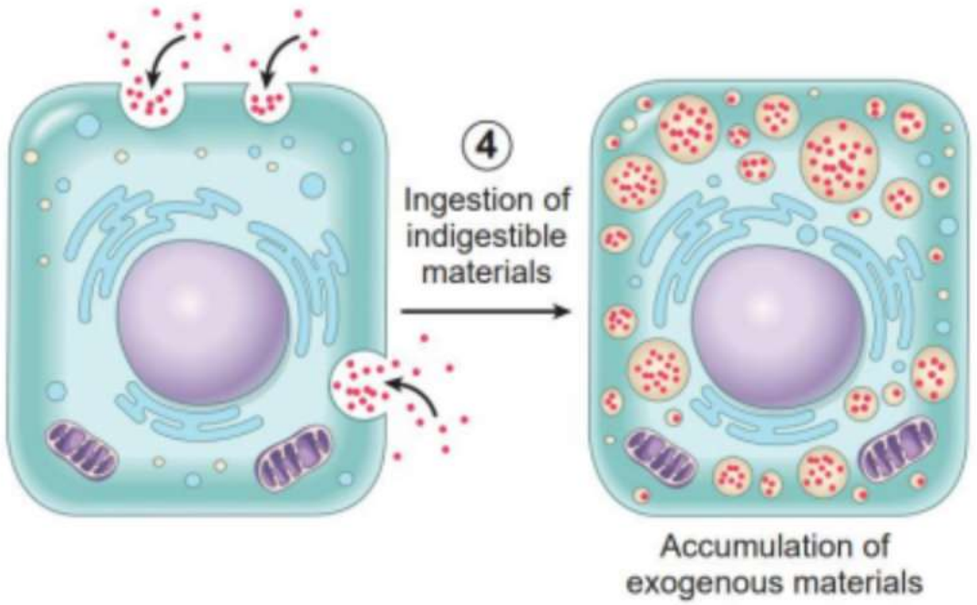


- **1. Inadequate removal of normal substances**
  - Defect in packaging/transport
  - Example: **fatty change (steatosis) in liver**
- **2. Accumulation of abnormal endogenous substances**
  - Defect in folding, packaging, transport, or secretion
  - Example: **mutant α_1-antitrypsin → protein accumulation**

Mechanisms of Intracellular Accumulation

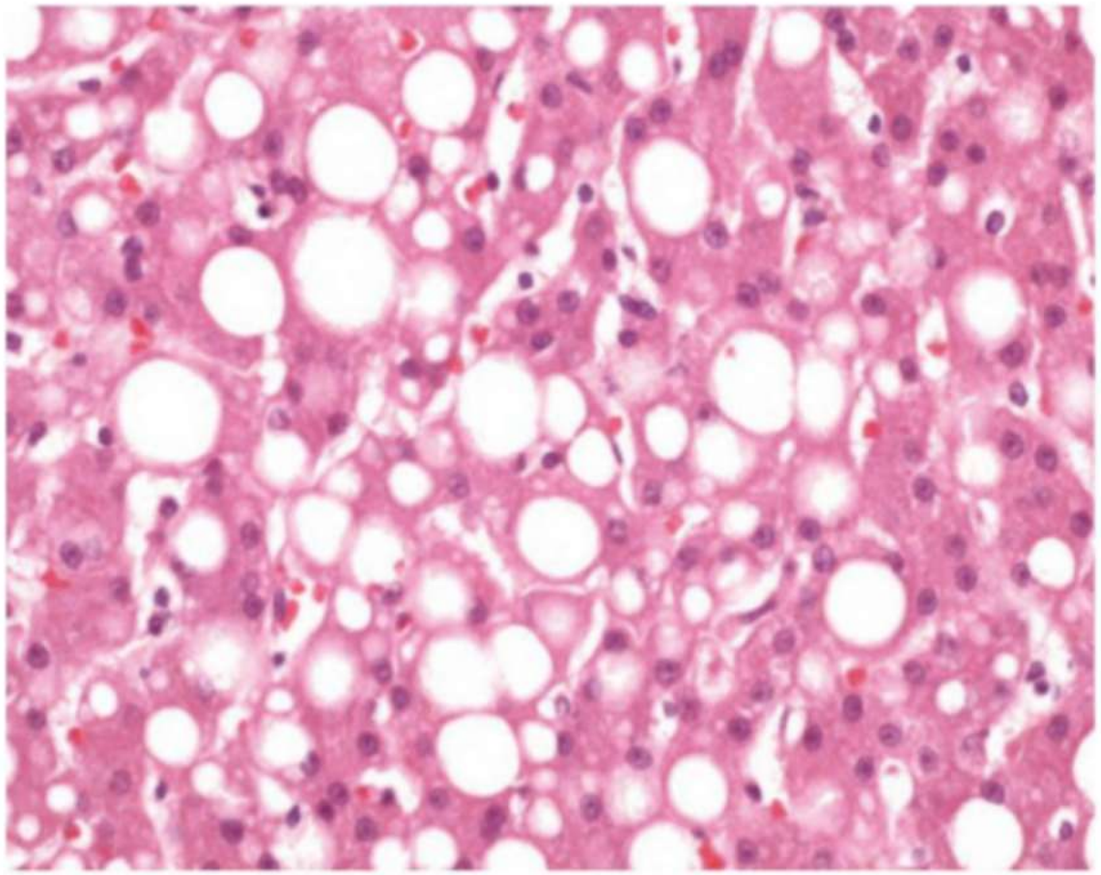


- **3. Failure to degrade endogenous substances**
- Usually due to **inherited enzyme deficiency**
- Typically, **lysosomal enzyme defects**
- Leads to **lysosomal storage diseases**



- **4. Accumulation of indigestible exogenous substances**
- The cell lacks enzymes to degrade them
- Or cannot transport them out
- Examples: **carbon (anthracosis), silica**

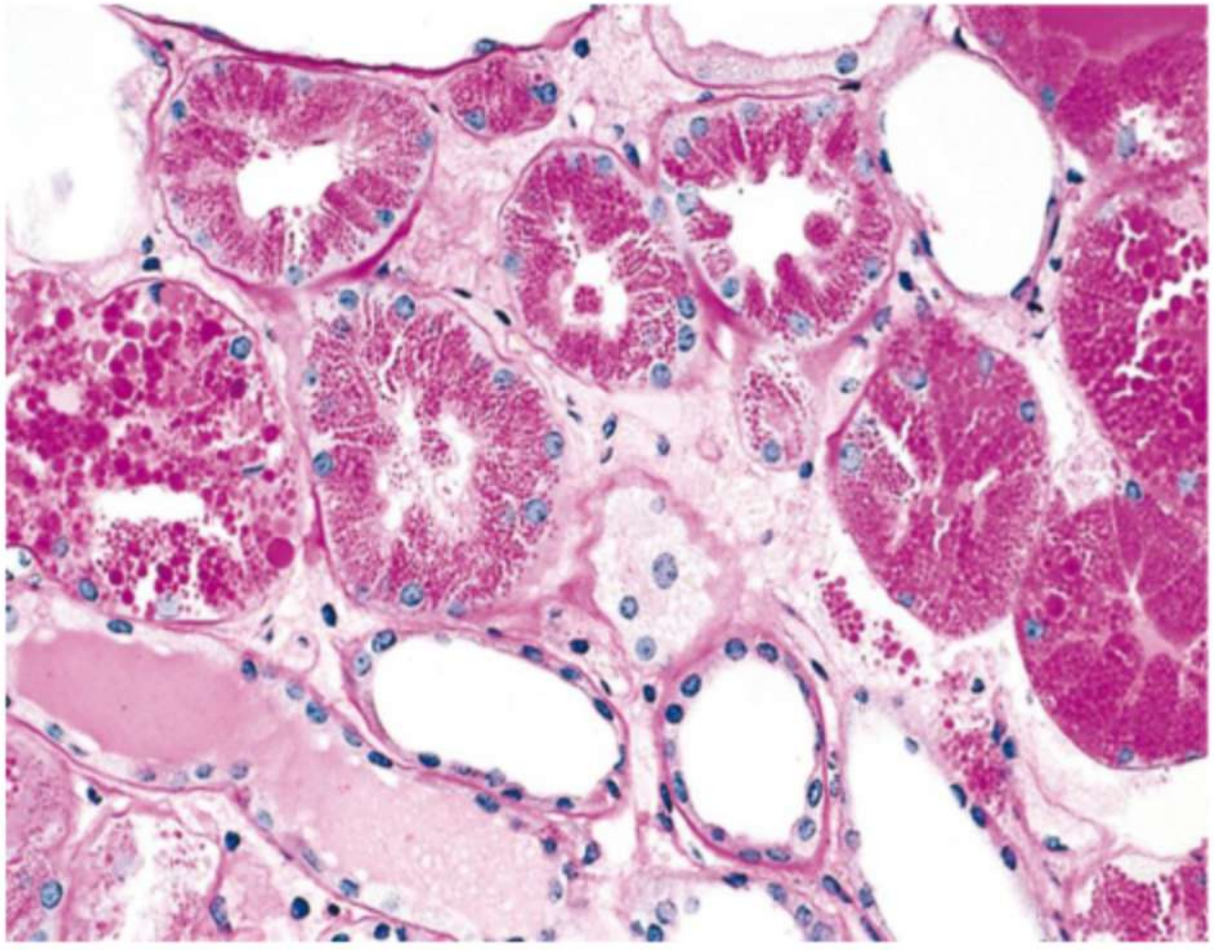
Lipid Accumulation (Steatosis)



- Lipids that can accumulate: triglycerides, cholesterol/cholesterol esters, phospholipids
- Triglyceride accumulation (fatty change) is most common
  - – typically in the liver, the main organ of fat metabolism
- Phospholipids contribute to myelin figures in necrotic cells
- Other examples:– Atherosclerosis (cholesterol in arterial intima)

Protein Accumulation

- Protein reabsorption droplets in the renal tubular epithelium in patients with proteinuria.



Intracellular accumulations of proteins usually appear as rounded, eosinophilic droplets, vacuoles, or aggregates in the cytoplasm

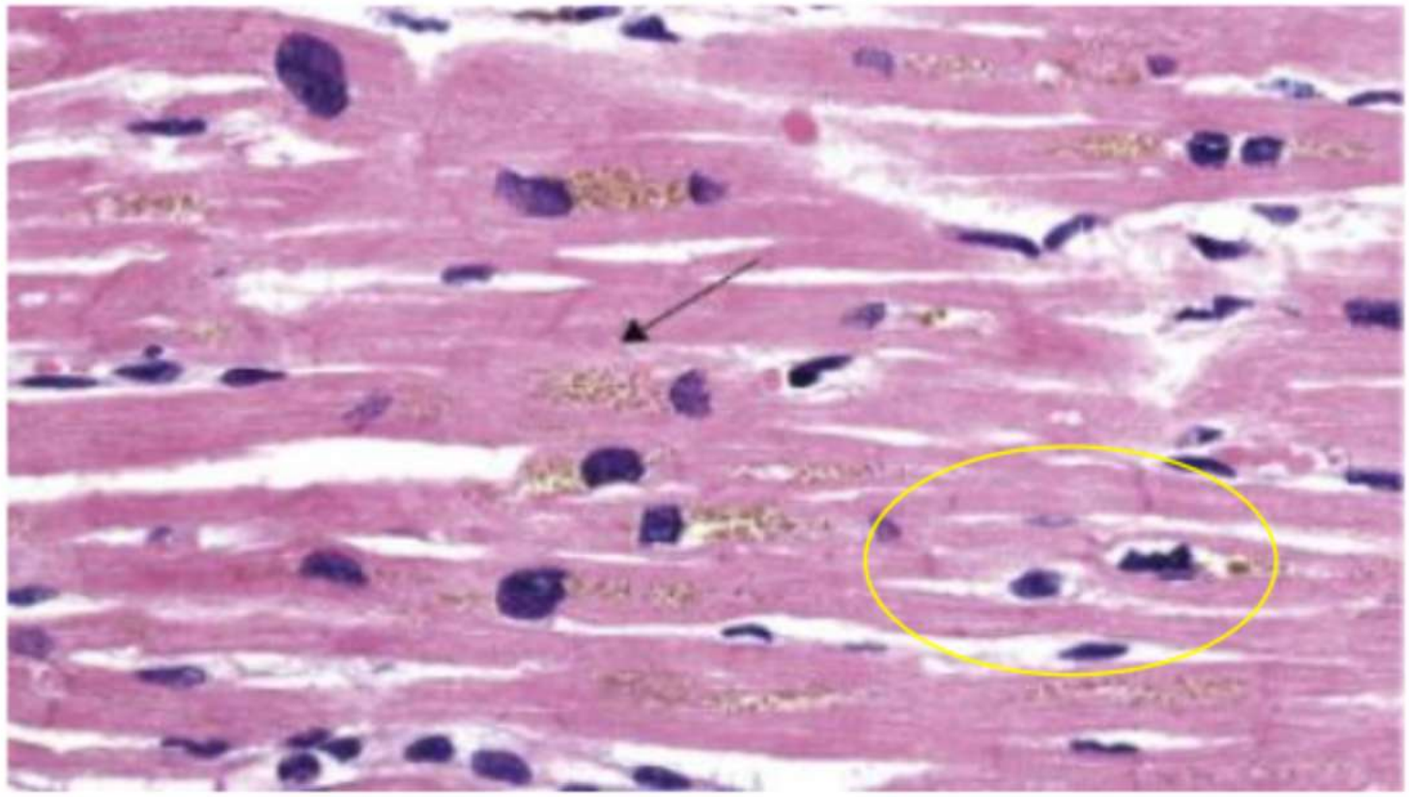
Pigments



- Pigments are coloured substances in cells
- Can be normal (e.g., melanin) or abnormal
- Can be **exogenous** or **endogenous**

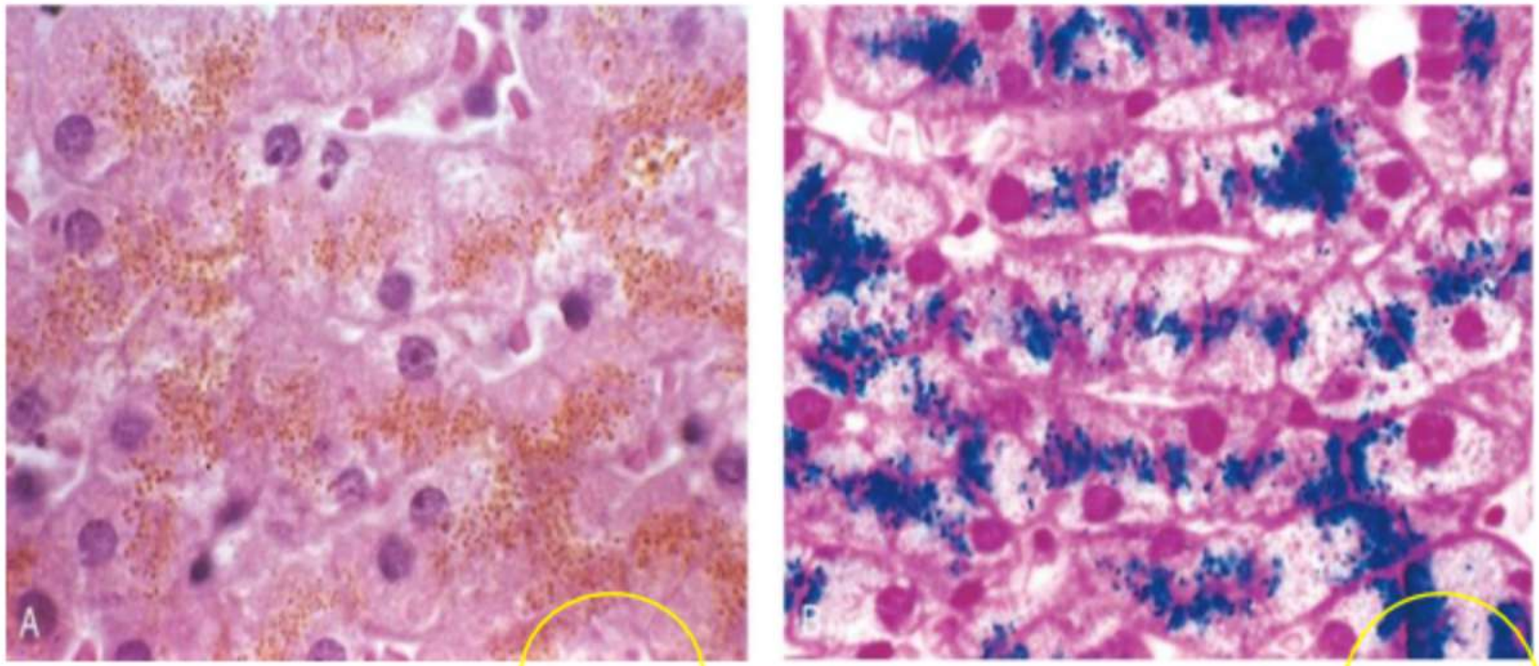
- **Exogenous pigments**
- **Carbon (coal dust)**
  → inhaled
  → taken up by macrophages
  → deposited in lungs & lymph nodes → *anthracosis*
- **Tattoo pigments**
  → pigment injected into dermis
  → phagocytosed by macrophages

Endogenous Pigments



- Lipofuscin granules in a cardiac myocyte shown by light microscopy (deposits indicated by arrows)
- **Lipofuscin is an insoluble pigment, also known as lipochrome or wear-and-tear pigment.** Lipofuscin is not injurious to the cell or its functions.

Endogenous Pigments



- Hemosiderin granules in liver cells. Left: H&E — golden-brown granular pigment Right: Prussian blue — iron stain (blue granules)
- **Hemosiderin granules represent aggregates of ferritin micelles.**
- **Derived from haemoglobin breakdown; yellow–brown, coarse granules.**
- **Indicates local or systemic iron excess (e.g., haemorrhage, haemochromatosis).**
- **Prussian blue stain confirms iron.**

Learning objectives

At the end of the teaching session, the student should be able to:

- Describe differences between hyperplasia, hypertrophy, atrophy and metaplasia.
- Explain causes and mechanism of cell injury.
- Describe reversible cell injury: fatty change, hydropic degeneration and briefly about intracellular accumulations.
- Describe different types of intracellular accumulations.